

Simulation & Modeling

BCE7026 — Semester VII, B.E. Computer Engineering

Purbanchal University · Full Marks 60 · Pass Marks 24 · 3:00 hrs

Complete notes, quiz answers and the past-question bank

8 chapters · 92 past questions (71 with model answers, 21 practice-only)

Theory weights: 8 + 6 + 6 + 6 + 6 + 12 + 6 + 10 marks

Contents

1. Concept of Simulation	8 marks	12 past questions
2. Monte Carlo Method	6 marks	8 past questions
3. Simulation of Continuous Systems	6 marks	10 past questions
4. Queuing System	6 marks	9 past questions
5. Verification & Validation	6 marks	5 past questions
6. Random Number Generation	12 marks	27 past questions
7. Analysis of Simulation Output	6 marks	6 past questions
8. Simulation Language	10 marks	15 past questions

Model answers are written for the questions that have appeared in real papers two or more times, and for every question in Chapters 5 and 6. Remaining questions are printed as practice questions so nothing is missing from the list.

1.1 Introduction to Simulation

Simulation is the imitation of the operation of a real-world process or system over time. It involves designing a model of a real system and conducting experiments with this model to understand its behavior or evaluate strategies for its operation.

Key Formula

Simulation = Model + Experimentation

The process of creating a model for simulation is known as *modeling*.

Purpose of Simulation:

- To reduce failure and meet specifications
- To prevent under and over-utilization of resources
- To eliminate unforeseen bottlenecks
- To optimize system performance

History of Simulation:

- **1940s:** Monte Carlo method developed by John Von Neumann (Manhattan Project — neutron scattering)
- **1960s:** First special-purpose languages (SIMSCRIPT by Harry Markowitz at RAND Corporation)
- **1970s:** Mathematical foundations of simulation research initiated
- **1980s:** PC-based simulation software, GUI, and object-oriented programming
- **1990s:** Web-based simulation, simulation-based optimization, Markov chain methods

Difference between Modeling & Simulation

Modeling	Simulation
A model is a physical or digital product that represents a system of interest	A simulation is the process of using a model to study the behavior and performance of a system
A model is similar but simpler than the original system	Simulation can study existing or proposed characteristics
A good model is a trade-off between realism and simplicity	Purpose is to study characteristics by manipulating variables
Can be physical (boat model) or conceptual (mathematical model)	Simulating is the act of using a model for simulation
Modeling is the act of building models	Results help in decision making

1.2 The System

The term **system** is derived from Greek word "*systema*" which means an organized relationship of functional units or components. The interaction and interdependence among the objects to fulfill a particular objective is called a system.

Examples: Transportation System, Telephone System, Production System, Accounting System, Banking System

System Concepts

System Boundary: Everything within the system is *endogenous*; everything outside is *exogenous*.

System Environment: The external conditions that affect the system.

Components of a System

Component	Definition	Example (Banking System)
Entity	An object of interest in the system	Client, Bank Teller, ATM, Account
Attribute	A property of an entity	Name, Account Number, Balance, Teller ID
Activity	A process causing changes in a system	Opening account, Depositing amount, Processing loan
State	Collection of variables necessary to describe the system at any time	Number of customers waiting, Number of busy tellers
Event	An instantaneous occurrence that changes the state	Arrival of customer, Completion of service

Types of Events

- **Endogenous Events:** Events occurring within the system (e.g., completion of service)
- **Exogenous Events:** Events occurring in the environment that affect the system (e.g., arrival of customer)

Example: Banking System

Entities: Client, Bank Teller, ATM, Account

Attributes: Name, Transaction History, Account Number, Balance, Teller ID, Service Time

Activities: Opening account, Depositing, Transfer, Processing loan

Events: Customer arrival (exogenous), Service completion (endogenous)

State Variables: Number of registered clients, Customers waiting in line, Busy tellers

Types of Systems

- **Endogenous System:** Activities and events occurring within a system (e.g., withdrawing cash)
- **Exogenous System:** Activities and events in the environment that affect the system (e.g., arrival of customer, strike)
- **Closed System:** No exogenous activity/events (e.g., water in an insulated flask)
- **Open System:** Has exogenous activities/events (e.g., bank system)

1.3 Continuous vs Discrete Systems

Feature	Continuous System	Discrete System
State changes	Predominantly smooth/continuous	Predominantly discontinuous (at discrete points)
Mathematical model	Differential equations	Event-driven logic
Example	Head of water behind a dam, aircraft movement	Customer arriving at bank, inventory changes
Simulation clock	Advances continuously	Advances in steps/events
Simulation language	CSSL, MATLAB/Simulink	GPSS, SIMSCRIPT, Arena

1.4 Deterministic vs Stochastic Systems

Feature	Deterministic System	Stochastic System
Random elements	No random elements, no uncertainty	One or more random elements
Output	Single & predictable output for given inputs	Range of possible outcomes, different each run
Example	$P(t) = P_0 e^{rt}$ (Population growth), $F = ma$ (Newton's law)	Stock market model (depends on investor behavior, economic indicators)

1.5 Real-Time Simulation

Real-time simulation means the model runs at the **same speed** as the actual system. The simulation clock advances in real-world time.

Applications

Flight simulators, military training, vehicle testing, hardware-in-the-loop simulations, medical procedure training.

1.6 When to Use Simulation

When Simulation IS Appropriate:

1. To address dynamic processes
2. To analyze the effect of alterations on model behavior
3. To determine the most important variables by changing inputs and observing outputs
4. To verify the analytical solution
5. To experiment new designs before actual implementation

When Simulation is NOT Appropriate:

1. When the problem can be solved by common sense
2. If the problem can be solved analytically
3. Easier to perform direct implementation
4. Cost exceeds the savings
5. Resources and time are not available
6. No data is available

1.7 Types of Simulation Models

Classification Basis	Types
Nature of output	Deterministic / Stochastic
Time change	Continuous / Discrete
Time dependency	Static / Dynamic
Scale	Physical / Mathematical
Methodology	Monte Carlo / Discrete Event / Continuous

Model Classification Tree

Model → **Physical | Mathematical**

Physical: Scale-Up (for tiny particles like molecules), Scale-Down (for large systems like bridges), Full-Scale (actual size, e.g., crash testing)

Mathematical → **Static | Dynamic**

Static: Snapshot at a particular moment (e.g., supply-demand models)

Dynamic: Evolves over time, represented by differential equations (e.g., population growth)

Mathematical → **Analytical | Numerical**

Analytical: Exact mathematical expressions (may be difficult for complex systems)

Numerical: Computational methods to approximate solutions (System Simulation)

1.8 Steps in Simulation Study

12 Steps in Simulation Study (Very Important for Exams)

This topic is asked in almost every exam paper!

1. **Problem Formulation:** Clearly state the problem
2. **Setting Objectives & Overall Plan:** How to approach the problem
3. **Model Conceptualization:** Establish a reasonable model
4. **Data Collection:** Collect necessary data (arrival process, service discipline, service rate)
5. **Model Translation:** Convert the model into a programming language
6. **Verification:** Verify whether the model works properly or not
7. **Validation:** Check if the system accurately represents the real system
8. **Experimental Design:** How many runs, how long, what kind of inputs
9. **Production Run & Analysis:** Actual running of the simulation model, collect & analyze output
10. **More Runs:** Repeat the experiment if necessary
11. **Documentation & Reporting:** Document the results & report to stakeholders
12. **Implementation:** Implement the actual model

Flowchart of the simulation study

The twelve steps in one picture — draw this in the answer, then label the two loops. The *verification and validation* loop is what makes it a simulation study and not a straight line, and the *more runs* loop is what turns one experiment into a statistically usable result.

1 Problem formulation 2 Setting objectives & overall plan 3 Model conceptualization 4 Data collection 5 Model translation 6 Verification (does it work?) 7 Validation (is it the right model?) 8 Experimental design 9 Production run & analysis 10 More runs needed? 11 Documentation & reporting 12 Implementation revise the model re-run with new seeds Fig 1.1 — Steps in a simulation study. Colour marks the four phases of §1.9: problem definition (steps 1–3), translation and data (4–5), verification, validation and design (6–8), analysis to implementation (9–12).

1.9 Phases of Simulation Study

Phase	Name	Steps Involved
Phase 1	Problem Definition & Conceptual Modeling	Steps 1, 2, 3
Phase 2	Model Translation & Data Collection	Steps 4, 5
Phase 3	Verification, Validation & Experimental Design	Steps 6, 7, 8
Phase 4	Analysis & Interpretation	Steps 9, 10, 11, 12

1.10 Advantages of Simulation

1. Useful for **sensitivity analysis** of complex systems
2. Suitable for analyzing complex real-life problems that cannot be solved using quantitative methods
3. If remaining tools cannot be used, simulation can be the option
4. Can be used as a tool for **testing and trying new policies** & decision rules for operating the system

1.11 Limitations of Simulation

1. Building a model requires **art that is learned through experience**
2. In some cases, results may be **complex to interpret**
3. It may be **time consuming and expensive**

1.12 Areas of Application

Area	Applications
Manufacturing	Process optimization, design testing, employee training
Healthcare	Medical procedure training, patient flow management, system design
Transportation	Traffic management, vehicle design, route optimization
Finance	Market value prediction, risk analysis, portfolio management
Military	Combat training, mission planning, strategies development

Past questions (12 of 12)

2025 M · 4 marks · repeated 4x

Explain the steps in a simulation study with a flowchart.

Answer

The 12 steps form a flowchart with feedback loops:

1. **Problem Formulation** → 2. **Setting Objectives & Plan** → 3. **Model Conceptualization**
2. **Data Collection** → 5. **Model Translation**
3. **Verification** (No → go back to step 5) → 7. **Validation** (No → go back to step 3)
4. **Experimental Design** → 9. **Production Run & Analysis**
5. **More Runs?** (Yes → go back to step 8) → 11. **Documentation & Reporting**
6. **Implementation**

The flowchart shows that verification failure leads back to model translation, validation failure leads back to model conceptualization, and the need for more runs leads back to experimental design. This iterative nature ensures the simulation model accurately represents the real system.

2010 F · 10 marks · repeated 4x

Discuss real-time simulation with an example. Mention purposes and limitations of simulation.

Answer

Real-Time Simulation: The model runs at the same speed as the actual system. The simulation clock advances in real-world time.

Example — Flight Simulator: A flight simulator creates a virtual cockpit with realistic controls, instruments, and visual displays. When a pilot moves the control stick, the simulated aircraft responds in real-time. Weather conditions, engine failures, and emergencies can be simulated to train pilots safely. The simulation must process inputs and update outputs at the same rate as a real aircraft.

Purposes of Simulation

1. To reduce failure and meet specifications
2. To prevent under/over-utilization of resources
3. To eliminate unforeseen bottlenecks
4. To optimize system performance
5. To test new designs before implementation

Limitations

1. Requires art and experience to build good models
2. Results may be difficult to interpret
3. Can be time consuming and expensive
4. May not produce exact answers — only estimates

2019 F · 5 marks · repeated 3x

Explain the concept of System. Differentiate between Continuous and Discrete, Deterministic and Stochastic system with example.

Answer

System: A system (from Greek 'systema') is a group of objects joined together in regular interaction or interdependence toward the accomplishment of some purpose. It consists of entities, attributes, activities, events, and state variables.

Example: A banking system consists of customers, tellers, ATMs, and accounts — all interacting to provide banking services.

Continuous vs Discrete System

Continuous System	Discrete System
State changes are predominantly smooth	State changes are predominantly discontinuous
Modeled by differential equations	Modeled by event-driven logic
Example: Head of water behind a dam changes smoothly	Example: Number of customers in bank changes at discrete points

Deterministic vs Stochastic System

Deterministic System	Stochastic System
No random elements, no uncertainty	One or more random elements
Same output for same inputs every time	Different output for same inputs each run
Example: $P(t) = P_0 e^{rt}$ (population growth), $F = ma$	Example: Stock market model (depends on investor behavior, economic indicators)

2014 F · 10 marks · repeated 3x

Explain the ways to study a system. What are the criteria of classifying Simulation models? Brief it with suitable example.

Answer

Ways to Study a System:

1. **Experiment with actual system:** Directly test the real system (costly, may be dangerous)
2. **Experiment with model of system:** Build a model and experiment (safer, cheaper)

Models can be: **Physical models** (tangible 3D representations) or **Mathematical models** (equations and algorithms).

Mathematical models can be solved **analytically** (exact solutions) or by **simulation** (numerical experimentation).

Classification Criteria for Simulation Models

Criterion	Types	Example
Nature of output	Deterministic vs Stochastic	$F=ma$ (deterministic) vs Stock market (stochastic)
Time change	Continuous vs Discrete	Water dam (continuous) vs Bank queue (discrete)
Time dependency	Static vs Dynamic	Supply-demand (static) vs Population growth (dynamic)
Scale	Physical vs Mathematical	Ship model (physical) vs Equation model (mathematical)

2011 F · 10 marks · repeated 3x

Define simulation. Explain the ways to study a system. Point out the areas where Simulation and Modeling can be used.

Answer

Simulation is the imitation of real-world process or system over time. It involves creating a model and conducting experiments to understand behavior or evaluate strategies.

Ways to Study a System

1. **Experiment with actual system:** Directly test (e.g., test-drive a car)

Experiment with model:

- **Physical model:** Scale models (ship, building)
- **Mathematical model:** Analytical solution or Simulation

Application Areas

1. **Manufacturing:** Process optimization, design testing, employee training
2. **Healthcare:** Medical procedure training, patient flow management
3. **Transportation:** Traffic management, vehicle design, route optimization
4. **Finance:** Market value prediction, risk analysis, portfolio management
5. **Military:** Combat training, mission planning, strategy development

6. **Telecommunications:** Network design, capacity planning

2011 C · 3+7 marks · repeated 3x

Explain real time simulation with example. Explain the type of simulation models

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 10 marks · repeated 2x

What do you mean by simulation and modeling? Explain the steps in the simulation study.

Answer

Simulation is the imitation of real-world process or system over time. It involves designing a model and conducting experiments to understand system behavior.

Modeling is the process of representing a model which includes its construction and working. The model is similar to the real system and helps analyze and predict the effect of changes.

12 Steps in Simulation Study:

1. **Problem Formulation:** Clearly define the problem
2. **Setting Objectives:** Determine goals and overall plan
3. **Model Conceptualization:** Build abstract model — start simple, add complexity
4. **Data Collection:** Gather input data (arrival rates, service times)
5. **Model Translation:** Code the model in simulation language
6. **Verification:** Does the code do what we intended?
7. **Validation:** Does the model accurately represent reality?
8. **Experimental Design:** Plan experiments — parameters to vary, replications
9. **Production Runs & Analysis:** Execute and analyze output statistically
10. **More Runs?** If more accuracy needed, run again
11. **Documentation & Reporting:** Document everything for decision-makers

12. **Implementation:** Apply the simulation findings

These 12 steps are iterative — you may go back to earlier steps if problems are found during verification or validation.

2012 C · 10 marks · repeated 2x

Define real time simulation. Mention the advantages and disadvantages of simulation.

Answer

Real-Time Simulation: A simulation where the model runs at the same speed as the actual system. The simulation clock advances in real-world time. It is used when the simulation must interact with real-world components or human operators in real-time.

Examples: Flight simulators for pilot training, military combat training simulators, vehicle testing (hardware-in-the-loop), medical procedure simulators.

Advantages of Simulation

1. Useful for sensitivity analysis of complex systems
2. Suitable for analyzing complex problems that cannot be solved analytically
3. Can be used when other tools are not applicable
4. Tool for testing new policies & decision rules without disrupting the real system
5. Can compress or expand time — observe years of operation in minutes
6. Can identify bottlenecks and inefficiencies in system design

Disadvantages of Simulation

1. Building a model requires art that is learned through experience
2. Results may be complex to interpret
3. May be time consuming and expensive
4. Each run produces only estimates, not exact answers

2025 M · 2 marks

Define system simulation and give one example.

Answer

System Simulation is the imitation of the operation of a real-world process or system over time. It involves designing a model of a real system and conducting experiments with this model to understand its behavior or evaluate strategies for its operation.

Example: A flight simulator used to train pilots. The simulator models the aircraft, weather conditions, and airport environment, allowing pilots to practice without risk. The simulation runs in real-time, responding to pilot inputs just like an actual aircraft would.

2011 C · 5 marks

Differentiate between discrete system simulation and continuous system simulation

Practice question — model answer not written yet. Attempt it from the chapter notes.

2010 C · 10 marks

What is system modeling? What are the types of system models? Explain dynamic physical model with an example.

Answer

System Modeling: The process of representing a model which includes its construction and working. The model is similar to the real system, helping analyze and predict effects of changes.

Types of System Models

1. Physical Model: Tangible 3D representation of a real system.

- **Scale-Down Model:** For large systems (bridges, airplanes, buildings)
- **Scale-Up Model:** For tiny particles (molecules, electrons, microstructures)
- **Full-Scale Model:** Actual size (e.g., airplane crash testing)

2. Mathematical Model: Uses equations, inequalities, and functions.

- **Static:** Does not consider changes over time (supply-demand)
- **Dynamic:** Describes how system evolves over time (population growth)

- **Analytical:** Exact mathematical expressions
- **Numerical:** Computational approximation methods

Dynamic Physical Model - Example

A **wind tunnel model** of an airplane is a dynamic physical model. The airplane model is placed in the wind tunnel, and air is blown at various speeds. The forces acting on the model change dynamically with airspeed, angle of attack, etc. Engineers observe how lift and drag change over time to optimize aircraft design.

2010 C · 3 marks

Differentiate between deterministic and stochastic simulation

Practice question — model answer not written yet. Attempt it from the chapter notes.

2.1 Monte Carlo Method

The **Monte Carlo method** is a computational technique that uses *random sampling* to obtain numerical results for problems that might be deterministic in principle. It was developed in the **1940s** by **John Von Neumann** and other physicists working on the Manhattan Project to study neutron scattering.

Basic Philosophy

Use **randomness** to approximate **deterministic** quantities. Generate random inputs → perform computation → aggregate results to get an estimate.

Steps in Monte Carlo Method

1. **Define the domain** of possible inputs
2. **Generate random inputs** from a probability distribution over the domain
3. **Perform a deterministic computation** on the inputs
4. **Aggregate the results** to get the final answer

Monte Carlo Integration

Used to estimate definite integrals using random numbers:

$$I \approx (b - a)/N \times \sum f(x_i) \text{ where } x_i \text{ are random numbers in } [a, b]$$

The actual value of the integral: $I = \int_a^b f(x)dx$

There are two Monte Carlo integration methods — learn both

They give different estimates for the same integral, so you must be able to recognise which one a question is asking for.

1. Sample-mean (average value) method — the one used for the cubic on this page. Generate R_i uniform on $[0,1]$, map them into $[a, b]$ with $x_i = a + (b - a)R_i$, evaluate f at each x_i and average:

$$I \approx (b - a)/N \times \sum f(x_i)$$

2. Hit-or-miss (rectangle) method — the one used in the class notes, worked on $\int_0^3 x^2 dx$ below. Enclose the area in a rectangle, scatter points (x, y) through it, and count how many land under the curve:

$$I \approx (b - a) \times (F(b) - F(a)) \times n/N$$

Method 1 needs one random number per sample; method 2 needs two (one for x , one for y). Method 1 is more accurate for the same N because it uses the function value directly instead of a yes/no decision.

Method 1 — Sample-Mean Integration

Estimate $\int_1^4 (x + 4)^3 dx$ by the sample-mean method $N = 6$

a1

b4

$f(x)(x + 4)^3$

N6

- Exact value by calculus: $\int_1^4 (x + 4)^3 dx = [(x + 4)^4 / 4]_1^4 = (8^4 - 5^4)/4 = (4096 - 625)/4 = 867.75$ Compute this first — you cannot report an error without it.
- Map each uniform R_i into the interval: $x_i = a + (b - a)R_i = 1 + 3R_i$ i R_i $x_i = 1 + 3R_i$ $f(x_i) = (x_i + 4)^3$

1	0.12	1.36	153.99	2	0.34	2.02	218.17	3	0.56	2.68	298.08	4	0.78	3.34	395.45	5	0.90	3.70	456.53	6	0.47	2.41	263.37
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 $\sum f(x_i) = 1785.59$ $x_i = 1 + 3R_i$ spreads the six values evenly across $[1, 4]$ — this is the only place the interval enters the calculation.
- $I \approx (b - a)/N \times \sum f(x_i) = 3/6 \times 1785.59 = 0.5 \times 1785.59 = 892.80$
- Error = |estimate - exact| = |892.80 - 867.75| = 25.05 Relative error = 25.05 / 867.75 $\times$ 100% = 2.89%

$I \approx 892.80$ exact 867.75 · error 25.05 (2.89%)

Method 2 — Hit-or-Miss (Rectangle) Integration

This is the method used in the class notes, and it is the one most exam questions expect. The area is boxed into a rectangle, points are scattered through the rectangle, and a point counts if it lands **below the curve**.

Estimate $\int_0^3 x^2 dx$ by the hit-or-miss method Lesson example · $N = 10$

a0

b3

$f(x)x^2$

N10

- $y = f(x) = x^2$ At $x = a = 0$: $f(0) = 0$ At $x = b = 3$: $f(3) = 9$
- Boundaries: $0 < x < 3$ and $0 < y < 9$ Area of rectangle = $(b - a) \times (F(b) - F(a)) = (3 - 0) \times (9 - 0) = 27$
- Decision rule for one point (x, y) : if $y - F(x) \leq 0$ the point is INSIDE (under the curve) otherwise the point is OUTSIDE
- Scatter the points and test each one: S.N. x y $y - x^2$ Remark

1	0	0	0.000	In	2	1.79	5.65	2.446	Out
3	0.87	3.16	2.403	Out	4	1.65	7.18	4.457	Out
5	0.842	4.219	3.510	Out	6	2.612	6.693	-0.130	In
7	1.803	1.399	-1.852	In	8	2.432	8.53	2.615	Out
9	2.124	2.184	-2.327	In	10	0.667	2.921	2.476	Out

 $n = 4$ points inside, $N = 10$ x must be drawn in $[0, 3]$ and y in $[0, 9]$ — not both in $[0, 1]$. A point exactly on the curve counts as inside, which is why $(0, 0)$ with $y - x^2 = 0$ is an "In".
- $I \approx \text{Area of rectangle} \times n/N = 27 \times 4/10 = 10.8$
- Exact value: $\int_0^3 x^2 dx = [x^3/3]_0^3 = 27/3 = 9$ Error = |10.8 - 9| = 1.8 Relative error = 1.8/9 $\times$ 100% = 20%

$I \approx 10.8$ exact 9 · error 1.8 (20%)

The count n is where the marks are lost

Everything else in this method is mechanical, so **tabulate the n/N count carefully** — one point classified the wrong way changes the answer by 2.7 (= 27/10) here. Check that the In and Out counts add up to N before you compute the integral.

Error and Convergence of Monte Carlo Estimates

The Monte Carlo estimate is a random variable, so it carries a **standard error** rather than a fixed error:

$$SE = (b - a) \times s / \sqrt{N} \quad \text{where } s^2 = (1/(N-1)) \times \sum (f(x_i) - \bar{f})^2$$

Two consequences are worth remembering:

- **Error falls only as $1/\sqrt{N}$** — to halve the error you must take four times as many samples, and to get one extra decimal digit you need 100 times as many.
- The estimate is **unbiased**: $E[\text{estimate}] = I$, so repeating the whole experiment many times and averaging the results converges to the true value (Law of Large Numbers).
- A **confidence interval** can be quoted, e.g. $I \approx \text{estimate} \pm 1.96 \times SE$ at 95% confidence.

Variance Reduction

Because plain Monte Carlo converges slowly, variance-reduction techniques are used to get a better answer from the same number of samples: **antithetic variates** (pair every R with $1 - R$), **stratified / importance sampling** (sample more densely where the function contributes most), and **control variates** (use a correlated quantity whose exact value is known).

Estimating π using Monte Carlo

One of the classic applications of Monte Carlo method:

1. Consider a circle $x^2 + y^2 = r^2$ inscribed in a square of side $2r$
2. Area of circle = πr^2 , Area of square = $(2r)^2 = 4r^2$
3. Ratio = $\pi r^2 / 4r^2 = \pi/4$
4. Generate random points (x, y) uniformly in the square
5. Count how many fall inside the circle ($x^2 + y^2 \leq r^2$)

$$\pi \approx 4 \times (\text{Points inside circle}) / (\text{Total points})$$

Two sampling conventions — both give $\pi \approx 4n/N$

First quadrant only (what the class notes use, written " $X = \text{Rand\#}(4)$, $Y = \text{Rand\#}(4)$ "): points are drawn in $[0, r]^2$, so the quarter-circle sits in a square of side r . The ratio is $(\pi r^2/4) / r^2 = \pi/4$.

Full circle (used in the worked numerical below): points are drawn in $[-r, r]^2$, so the circle sits in a square of side $2r$. The ratio is $\pi r^2 / (2r)^2 = \pi/4$.

Both ratios are $\pi/4$, so both give $\pi \approx 4 \times (\text{points inside}) / (\text{total points})$. Pick one convention, say which you used, and map your random numbers to the right range — drawing x in $[0, 4]$ and y in $[-4, 4]$ mixes the two and will be marked wrong.

Estimate π from $x^2 + y^2 = 16$, full circle = $4 \cdot N = 10$

Radius $r=4$

Square $[-4, 4]^2$, area 64

Circle $x^2 + y^2 \leq 16$, area 16π

$N=10$

- $\pi r^2 / (2r)^2 = 16\pi / 64 = \pi/4$ $\pi \approx 4 \times (\text{points inside circle}) / (\text{total points})$
- Map each pair of random numbers into the square: $x = -4 + 8R$, $y = -4 + 8R$.
P R x y $x^2 + y^2 \leq 16$?
1 0.15 0.72 -2.80 1.76 10.94 Yes
2 0.63 0.31 1.04 -1.52 3.39 Yes
3 0.88 0.95 3.04 3.60 22.20 No
4 0.42 0.07 -0.64 -3.44 12.24 Yes
5 0.77 0.55 2.16 0.40 4.83 Yes
6 0.24 0.83 -2.08 2.64 11.30 Yes
7 0.95 0.36 3.60 -1.12 14.21 Yes
8 0.51 0.66 0.08 1.28 1.64 Yes
9 0.09 0.19 -3.28 -2.48 16.91 No
10 0.70 0.44 1.60 -0.48 2.79 Yes
points inside = 8 of 10
 $x^2 + y^2 = 14.21 < 16$ is inside; $x^2 + y^2 = 16.91 > 16$ is outside — the comparison is against $r^2 = 16$, not against $r = 4$.
- $\pi \approx 4 \times 8/10 = 3.2$
- Error = $|3.2 - 3.14159| = 0.0584$ Relative error = $0.0584 / 3.14159 \times 100\% = 1.86\%$ Only 10 points are used so the arithmetic stays hand-workable. Because error falls as $1/\sqrt{N}$, 10 000 points typically give π to two or three decimal places.

$\pi \approx 3.2$ error **0.0584 (1.86%)**

Estimate π from $x^2 + y^2 \leq 36$ — the class numerical

Radius $r=6$

Square x, y in $[0, 6]$, area 36

Quarter circle area 9π

$N=15$

- Quarter circle in a square of side r : $(\pi r^2/4) / r^2 = \pi/4 \rightarrow \pi \approx 4n/N$ Written in the notes as $X = \text{Rand}\#(6)$, $Y = \text{Rand}\#(6)$ — both coordinates in $[0, 6]$. With $r = 6$ the point is inside when $x^2 + y^2 \leq 36$.
- S.N. x y $x^2 + y^2 \leq 36$ Remark
1 0 0 -36.000 In
2 4.342 2.165 -12.460 In
3 5.601 3.005 4.401 Out
4 1.668 0.967 -32.283 In
5 4.644 3.047 -5.149 In
6 3.005 4.974 -2.229 In
7 5.098 4.234 7.916 Out
8 1.664 4.339 -14.404 In
9 4.032 0.474 -19.518 In
10 3.317 5.963 10.560 Out
11 3.606 1.468 -20.842 In
12 3.937 2.874 -12.240 In
13 0.301 5.855 -1.628 In
14 3.418 5.646 7.560 Out
15 0.689 0.915 -34.688 In
Points 1–15 are all in the first quadrant, so all the x and y values are positive. A point is inside when $x^2 + y^2 \leq 36$, i.e. when $x^2 + y^2 \leq 36$.
- $n = 11$ points inside, $N = 15$ Check: $11 \text{ In} + 4 \text{ Out} = 15 = N$ ✓
- $\pi \approx 4 \times n/N = 4 \times 11/15 = 44/15 = 2.933$
- Taking the true value as $\pi = 3.14$: Error $E = |\pi_{\text{actual}} - \pi_{\text{calculated}}| = |3.14 - 2.9333| = 0.2067$ $E\% = E/\pi_{\text{actual}} \times 100\% = 0.2067/3.14 \times 100\% = 6.58\%$

$\pi \approx 2.933$ error **0.2067 (6.58%)**

Compare the two estimates honestly

Ten points gave $\pi \approx 3.2$ (1.86% error) and fifteen points gave $\pi \approx 2.933$ (6.58% error). More points did *not* give a better answer here — Monte Carlo error is **random**, not monotonic. What the $1/\sqrt{N}$ rule promises is that the error *tends* to shrink, and that to be confident of halving it you need four times as many points. Never claim a Monte Carlo estimate is better just because N is larger.

Monte Carlo for Business Simulation

Monte Carlo can simulate real business scenarios like inventory management, demand forecasting, and queuing problems by assigning random number ranges to probability distributions.

Vegetable Shopkeeper — the method

Given probability distributions for customers/day and kg/customer:

1. Assign random number ranges to each probability (cumulative probability $\times$ 100)
2. Generate random numbers for each day
3. Map random numbers to number of customers
4. For each customer, generate another random number for kg ordered
5. Calculate total demand per day

The ranges must tile 00–99 with no gap and no overlap, and the last cumulative probability must reach exactly 1.00 — otherwise some random numbers map to nothing.

Green-vegetable ordering for a 2-day simulationDemand & inventory

Customers/day8, 10, 12, 14

p0.35, 0.30, 0.25, 0.10

Kg/customer1, 2, 3, 4

p0.2, 0.4, 0.3, 0.1

1. Build the random-number ranges from the cumulative probabilities.

Customers/day	Probability	Cumulative	RN range
8	0.35	0.35	00–34
10	0.30	0.65	35–64
12	0.25	0.90	65–89
14	0.10	1.00	90–99

Kg ordered	Probability	Cumulative	RN range
1	0.2	0.2	00–19
2	0.4	0.6	20–59
3	0.3	0.9	60–89

4	0.1	1.0	90–99
---	-----	-----	-------

Simulate. One random number per day sets the customer count; then **one random number per individual customer** sets the quantity ordered.

Day	RN for customers	Customers	Individual RNs for kg	Kg per customer	Total demand
1	24	8	15, 47, 82, 66, 29, 91, 55, 38	1, 2, 3, 3, 2, 4, 2, 2	19 kg
2	71	12	22, 08, 63, 74, 45, 96, 31, 58, 88, 12, 40, 67	2, 1, 3, 3, 2, 4, 2, 2, 3, 1, 2, 3	28 kg

2. Day 1 cost 8 individual random numbers because 8 customers were drawn; day 2 cost 12. The number of samples per day is not fixed — it follows from the day's customer count.

3. Average demand = $(19 + 28)/2 = 23.5$ kg per day

Average demand 23.5 kg/day **buy about 24 kg each morning**

Note on method: taking the expected value ($E[\text{kg}] = 1 \times 0.2 + 2 \times 0.4 + 3 \times 0.3 + 4 \times 0.1 = 2.3$ kg) and multiplying by the number of customers gives the same average but hides the day-to-day variation. A proper simulation draws an individual random number for every customer, which is what makes it possible to report the spread of demand (and therefore the risk of running out) rather than only the mean.

Two days is also too short to draw a conclusion — with only two demands (19 and 28) the average could move a long way in either direction. State the number of replications a real study would need, and why.

2.2 Normally Distributed Random Numbers

Many natural phenomena follow a **normal (Gaussian) distribution**. To generate normally distributed random numbers from uniform random numbers:

Box-Muller Transform

Given two independent uniform random numbers R_1 and R_2 on $(0,1)$:

$$Z_1 = \sqrt{-2 \ln R_1} \times \cos(2\pi R_2) \\ Z_2 = \sqrt{-2 \ln R_1} \times \sin(2\pi R_2)$$

Z_1 and Z_2 are independent standard normal random variables (mean=0, variance=1).

To get normal with mean μ and standard deviation σ : $X = \mu + \sigma Z$

Central Limit Theorem Method

Sum of 12 uniform random numbers approximates normal distribution:

$$Z = (\sum_{i=1}^{12} R_i) - 6$$

where R_i are uniform random numbers on $(0,1)$. This gives approximately $N(0,1)$.

Why Summing 12 Uniform Numbers Works

Each uniform number R_i on $(0,1)$ has mean $1/2$ and variance $1/12$. For 12 independent values the sum therefore has

$$\text{mean} = 12 \times 1/2 = 6 \quad \text{variance} = 12 \times 1/12 = 1 \quad \text{so } Z = \sum R_i - 6 \text{ has mean } 0 \text{ and variance } 1$$

By the Central Limit Theorem the sum of many independent values is approximately normal, and 12 is the smallest count that makes the approximation both simple (an exact variance of 1) and acceptable in practice.

Box-Muller — two normal values from two uniforms

$R_1=0.35$

$R_2=0.60$

Target μ , σ 100, 15

- Radius term: $\sqrt{-2 \ln R_1} = \sqrt{-2 \ln 0.35} = \sqrt{-2 \times (-1.0498)} = \sqrt{2.0996} = 1.4490$ In of a number below 1 is negative, so $-2 \ln R_1$ is positive. If your radius term comes out negative or complex, R_1 is not in $(0, 1)$.
- Angle: $2\pi R_2 = 2\pi \times 0.60 = 3.7699$ rad $\cos(3.7699) = -0.8088$ $\sin(3.7699) = -0.5881$ Work in radians, not degrees. 3.7699 rad is just past π (3.1416), so both cos and sin are negative — the point is in the third quadrant.
- $Z_1 = 1.4490 \times \cos(3.7699) = 1.4490 \times (-0.8088) = -1.172$ $Z_2 = 1.4490 \times \sin(3.7699) = 1.4490 \times (-0.5881) = -0.852$ Z_1 and Z_2 are independent standard normal values (mean 0, variance 1).
- To a general normal $N(\mu, \sigma^2)$: $X = \mu + \sigma Z$ $X_1 = 100 + 15 \times (-1.172) = 100 - 17.58 = 82.42$ $X_2 = 100 + 15 \times (-0.852) = 100 - 12.78 = 87.22$

$Z_1, Z_2 = -1.172, -0.852 \rightarrow N(100, 15^2)$ 82.42, 87.22

Comparison of the Two Methods

Aspect	Box-Muller	Central Limit (sum of 12)
Accuracy	Exact normal distribution	Approximate — the tails are cut off
Speed	Slower — needs logarithm, sine and cosine	Very fast — additions only
Random numbers used	2 uniforms produce 2 normal values	12 uniforms produce 1 normal value
Range	Unbounded, correct extreme tails	Bounded to roughly [−6, +6]
Use when	Tail behaviour matters (risk analysis, rare events)	Speed matters and the tails are unimportant

Polar (Marsaglia) Method

A variant of Box-Muller that avoids the sine and cosine: generate u and v uniformly in $(-1, 1)$ and keep the pair only while $s = u^2 + v^2 < 1$, then

$Z_1 = u \times \sqrt{(-2 \ln s / s)}$ $Z_2 = v \times \sqrt{(-2 \ln s / s)}$

It discards some pairs (about 21% of them) but each accepted pair costs far less computation than a trigonometric Box-Muller pair.

2.3 Monte Carlo Method vs Stochastic Simulation

Feature	Monte Carlo Method	Stochastic Simulation
Nature	Uses random numbers to solve <i>deterministic</i> problems	Models systems with <i>inherent randomness</i>
Purpose	Numerical estimation of mathematical quantities	Study system behavior over time
Time element	Typically no time component	Evolves over simulated time
Examples	Computing π , evaluating integrals	Bank queuing, manufacturing process
Randomness role	Tool for computation	Inherent part of the system
System model	No system model needed	Requires detailed system model
Output	Single numerical estimate	Time-series of system states

Key Distinction

Monte Carlo uses randomness as a **computational tool** to solve deterministic problems. Stochastic simulation models systems that are **inherently random**.

Past questions (8 of 8)

2014 F · 5 marks · repeated 5x

Find the value of π by using Monte Carlo simulation using $x^2 + y^2 = 16$. Calculate error.

Answer

Given circle: $x^2 + y^2 = 16$ (radius $r = 4$)

Enclosing square: $[-4, 4] \times [-4, 4]$, side = 8

Area of circle = $\pi r^2 = 16\pi$

Area of square = $8^2 = 64$

Ratio = $16\pi/64 = \pi/4$

Therefore: $\pi = 4 \times (\text{Points inside circle}) / (\text{Total points})$

Simulation: Generate random points (x, y) where $-4 \leq x \leq 4$, $-4 \leq y \leq 4$

Suppose we generate 10 random points:

Point	x	y	x^2+y^2	Inside? (≤ 16)
1	1.2	3.5	13.69	Yes
2	-3.8	2.1	18.85	No
3	2.0	-1.5	6.25	Yes
4	-0.5	3.9	15.46	Yes
5	3.6	-1.8	16.20	No
6	-2.3	-2.7	12.58	Yes
7	0.8	1.2	2.08	Yes
8	-3.5	3.0	21.25	No
9	1.5	-3.0	11.25	Yes
10	-1.0	2.5	7.25	Yes

Points inside = 7, Total = 10

$$\pi \approx 4 \times 7/10 = 2.8$$

$$\text{Error} = |2.8 - 3.14159| = 0.34159$$

Note: With more points ($N \rightarrow \infty$), the estimate converges to actual π .

2019 F · 5 marks · repeated 4x

Discuss the basic philosophy of Monte-Carlo simulation.

Answer

The basic philosophy of Monte Carlo simulation is to use **randomness as a computational tool** to solve deterministic problems. Developed in the 1940s by John Von Neumann during the Manhattan Project.

Philosophy: Instead of solving a problem analytically (which may be impossible for complex problems), we:

1. **Define the domain** of possible inputs

2. **Generate random inputs** from probability distributions over the domain
3. **Perform deterministic computation** on each random input
4. **Aggregate results** — the average converges to the true answer as $N \rightarrow \infty$

Key insight: By the Law of Large Numbers, the average of many random samples converges to the expected value. So we can approximate deterministic quantities using random experiments.

Example: To find π , we randomly throw darts at a square containing a circle. The ratio of darts inside the circle to total darts approximates $\pi/4$.

2025 M · 4 marks · repeated 3x

Compare stochastic simulation vs. Monte Carlo method.

Answer

Feature	Monte Carlo Method	Stochastic Simulation
Nature	Uses random numbers to solve deterministic problems	Models systems with inherent randomness
Purpose	Numerical estimation of mathematical quantities	Study system behavior over time
Time element	Typically no time component	Evolves over simulated time
Examples	Computing π , evaluating integrals	Bank queuing, manufacturing process
Randomness role	Tool for computation	Inherent part of the system
System model	No system model needed	Requires detailed system model
Output	Single numerical estimate	Time-series of system states

2019 F · 10 marks

A vegetable shopkeeper is trying to determine how much green vegetable he has to purchase. No. of customers: 8(0.35), 10(0.30), 12(0.25), 14(0.10). Kg ordered: 1(0.2), 2(0.4), 3(0.3), 4(0.1). Simulate for 5 days.

Answer

Step 1: Assign random number ranges for customers/day

Customers/Day	Probability	Cumulative	Random # Range
8	0.35	0.35	00-34
10	0.30	0.65	35-64
12	0.25	0.90	65-89
14	0.10	1.00	90-99

Step 2: Assign random number ranges for kg/customer

Kg Ordered	Probability	Cumulative	Random # Range
1	0.2	0.2	00-19
2	0.4	0.6	20-59
3	0.3	0.9	60-89
4	0.1	1.0	90-99

Step 3: Simulate 5 days (using sample random numbers)

Day	RN	Customers	Avg Kg/Customer	Total Demand (Kg)
1	24	8	2.3 ($E[X] = 1 \times 0.2 + 2 \times 0.4 + 3 \times 0.3 + 4 \times 0.1 = 2.3$)	$8 \times 2.3 = 18.4$
2	61	10	2.3	$10 \times 2.3 = 23.0$
3	43	10	2.3	$10 \times 2.3 = 23.0$
4	78	12	2.3	$12 \times 2.3 = 27.6$
5	92	14	2.3	$14 \times 2.3 = 32.2$

Average daily demand = $(18.4 + 23 + 23 + 27.6 + 32.2) / 5 = 124.2 / 5 = 24.84$ kg

The shopkeeper should purchase approximately **25 kg** of vegetables per day.

Note: In a more detailed simulation, we would generate individual random numbers for EACH customer's order. The expected kg per customer is $E[X] = 1(0.2) + 2(0.4) + 3(0.3) + 4(0.1) = 2.3$ kg.

2012 C · 5 marks

Solve $\int \blacksquare \blacksquare dx$ using Monte Carlo Method

Practice question — model answer not written yet. Attempt it from the chapter notes.

2011 F · 5 marks

Calculate $\int_{-5}^8 (x+4)^3 dx$ using Monte Carlo simulation. Calculate error.

Answer

Actual value: $\int_{-5}^8 (x+4)^3 dx = [(x+4)^4/4]_{-5}^8 = (8^4 - 5^4)/4 = (4096 - 625)/4 = 867.75$

Monte Carlo estimation:

Formula: $I \approx (b-a)/N \times \sum f(x_i)$ where $a=1$, $b=4$, $f(x)=(x+4)^3$

Generate 5 random numbers R_i in $[0,1]$: 0.82, 0.15, 0.63, 0.45, 0.91

Transform to $[1,4]$: $x_i = 1 + 3 \times R_i$

i	R_i	$x_i = 1+3R_i$	$f(x_i) = (x_i+4)^3$
1	0.82	3.46	$(7.46)^3 = 415.16$
2	0.15	1.45	$(5.45)^3 = 161.88$
3	0.63	2.89	$(6.89)^3 = 327.08$
4	0.45	2.35	$(6.35)^3 = 256.05$
5	0.91	3.73	$(7.73)^3 = 461.89$

$\sum f(x_i) = 415.16 + 161.88 + 327.08 + 256.05 + 461.89 = 1622.06$

$I \approx (4-1)/5 \times 1622.06 = 3/5 \times 1622.06 = 973.24$

Error = $|973.24 - 867.75| = 105.49$

With more random samples, the error would decrease ($\text{error} \propto 1/\sqrt{N}$).

2010 C · 5 marks

Define normally distributed random number with example.

Answer

A **normally distributed random number** is a random number that follows the normal (Gaussian) distribution with a bell-shaped probability density function. It is characterized by mean (μ) and standard deviation (σ).

$$f(x) = (1/\sigma\sqrt{2\pi}) \times e^{-(x-\mu)^2/2\sigma^2}$$

Methods to generate:

1. Box-Muller Transform: Given $R_1, R_2 \sim \text{Uniform}(0,1)$:

$$Z_1 = \sqrt{-2 \ln R_1} \times \cos(2\pi R_2) \rightarrow \text{Standard Normal } N(0,1)$$

Example: $R_1 = 0.5, R_2 = 0.3$

$$Z_1 = \sqrt{-2 \times \ln(0.5)} \times \cos(2\pi \times 0.3) = \sqrt{1.386} \times \cos(1.885) = 1.177 \times (-0.309) = \mathbf{-0.364}$$

For normal with $\mu=100, \sigma=15$: $X = 100 + 15 \times (-0.364) = \mathbf{94.54}$

2. Central Limit Theorem: $Z = (\sum R_{i,j}) - 6$, where sum of 12 uniform numbers $\approx N(6,1)$

2010 C · 5 marks

Find the value of $\int_0^{\pi/2} \cos x \, dx$ using Monte-Carlo Method

Practice question — model answer not written yet. Attempt it from the chapter notes.

3.1 A Pure Pursuit Problem

A **pure pursuit problem** is a scenario where a vehicle or robot follows a pre-defined path by constantly adjusting its direction to minimize the distance between itself and the target. The target is *not aware* of the pursuer (unlike "hot pursuit").

Classic Example: Fighter Aircraft Chasing Bomber

A fighter aircraft sights an enemy bomber and flies directly towards it to catch up and destroy it. We must determine the course of attack and how long the fighter takes.

Assumptions

1. Both pursuer and bomber fly in the **same 2D plane**
2. The fighter speed is **constant**
3. The target path is **known**
4. Minimum distance to fire a missile = **10 units**
5. If target is not caught within given time → target **escapes**
6. Initial coordinates of fighter are known

Mathematical Formulation

At time $t=0$: Fighter at (0, 50), Target on x-axis at (90, 0)

$$\begin{aligned}d &= \sqrt{((x_b(t) - x_f(t))^2 + (y_b(t) - y_f(t))^2)} \\ \sin \theta &= (y_b(t) - y_f(t)) / d \\ \cos \theta &= (x_b(t) - x_f(t)) / d\end{aligned}$$

Next position of fighter at time (t+1):

$$\begin{aligned}x_f(t+1) &= x_f(t) + V_p \times \cos \theta \\ y_f(t+1) &= y_f(t) + V_p \times \sin \theta\end{aligned}$$

where V_p = velocity of fighter. The angle must match the axis: $\cos \theta$ is the fraction of the speed along x — it is $(x_b - x_f)/d$ — and $\sin \theta$ is the fraction along y , $(y_b - y_f)/d$. Swapping the two makes the fighter fly away from the bomber, which is the classic mistake in this question.

Pure pursuit — one time step, from the initial conditions given abovefighter (0, 50) · bomber (90, 0)

Fighter(x_f, y_f) = (0, 50)

Bomber(x_b, y_b) = (90, 0)

Stop rule**hit if $d \leq 10$ units**

1. $d = \sqrt{(x_b - x_f)^2 + (y_b - y_f)^2} = \sqrt{(90 - 0)^2 + (0 - 50)^2} = \sqrt{8100 + 2500} = \sqrt{10600} = 102.96$ units $d > 10$, so the pursuit continues — the first line of every answer is the distance check.
2. $\cos \theta = (x_b - x_f)/d = 90/102.96 = 0.8742$ (share of the speed along x) $\sin \theta = (y_b - y_f)/d = -50/102.96 = -0.4856$ (share along y)

3. $x_f(t+1) = 0 + V_p \times 0.8742 = +0.8742 V_p$ $y_f(t+1) = 50 + V_p \times (-0.4856) = 50 - 0.4856 V_p$ With $V_p = 20$ units per step this is (17.48, 40.29) — substitute the speed your question gives.
4. New distance: $\sqrt{[(90 - 0.8742V_p)^2 + (0 - (50 - 0.4856V_p))^2]}$ Recompute d each step; repeat until $d \leq 10$ (hit) or the allowed time runs out (escape). This is the whole simulation loop.

$d = 102.96$ units, still closing $\cos \theta = 0.8742 \cdot \sin \theta = -0.4856$

Algorithm

- If distance ≤ 10 units $\rightarrow$ **bomber is shot down**
- If the fighter has not closed to 10 units within the time allowed $\rightarrow$ **bomber escapes**
- Otherwise $\rightarrow$ recalculate θ from the new positions and take another step

If the target flies along a straight line, the problem can be solved **analytically**. If the path is curved, **simulation** is needed.

3.2 Continuous System Models

Continuous systems are those where system variables change continuously according to time. Quantities like water level, pressure, temperature, and flow rate are simulated by mathematical models through **differential equations**.

Advantages of Differential Equations for Continuous Simulation

1. **Dynamic Modeling:** Can model how quantities change with respect to time/space (temperature, population growth, circuits)
2. **Continuity:** Track quantities that change continuously (liquid flow, heat distribution)
3. **Predictive Power:** Once the differential equation is known, future behavior can be predicted
4. **Broad Application:** Engineering, physics, biology, economics

3.3 Analog Computer

An analog computer processes **continuous data/signals** using physical components such as electrical, mechanical, or hydraulic components.

Examples: OPAM (Operational Amplifier), Mechanical Integrators, Abacus, Analog Speedometer

Features

- Non-programmable
- Highly accurate but not highly precise
- Real-time processing/simulation
- Uses continuous signals (voltage or current)

3.4 Analog Methods — Components

The various components used in analog computers to simulate continuous systems:

Component	Symbol	Function
Summer (Adder)	Σ	Adds multiple input signals
Integrator	$\int$	Integrates signal over time
Scale Factor	K	Multiplies signal by a constant
Inverter	-1	Reverses the sign of signal

Automobile Suspension System

A key component connecting wheels to the vehicle body, absorbing shock on rough surfaces.

$$M\ddot{x} + D\dot{x} + Kx = Kf(t)$$

where M = mass, D = damping coefficient, K = spring constant, f(t) = force, x = displacement

Rearranging: $M\ddot{x} = Kf(t) - D\dot{x} - Kx$

Block diagram: 3 variables are added with a summer to produce $M\ddot{x}$ → scale by 1/M to get $\ddot{x}$ → integrator gives $\dot{x}$ → inverter gives $-\dot{x}$ → integrator gives $-x$ → inverter gives x.

Computer Model of Liver

Thyroxine enters blood → carried to liver → liver converts to iodine → iodine absorbed by bile

$$\begin{aligned} \dot{X}_1 &= K_{in}X_{in} - K_{out}X_1 \\ \dot{X}_2 &= K_{in}X_{in} - (K_{out} + K_{liver})X_2 \\ \dot{X}_3 &= K_{liver}X_2 \end{aligned}$$

3 integrators are used, each solving one differential equation. Necessary scaling is also applied.

3.5 Hybrid Simulation

When a system is **neither purely continuous nor purely discrete**, we need both analog and digital computers. A hybrid computer exhibits features of both.

Key Challenge

Hybrid simulation requires high-speed **DAC** (Digital to Analog Converter) and **ADC** (Analog to Digital Converter).

Example: An artificial satellite follows equations of motion (continuous) but also requires digital signal processing.

3.6 Feedback Systems

Systems in which the **output is sampled and fed back to the input**.

Example: Home Heating System

A thermostat-controlled furnace: The furnace heats the room (output = room temperature). The thermostat measures temperature and turns furnace ON/OFF based on whether temperature is below/above the setting.

Types of Feedback

Feature	Positive Feedback	Negative Feedback
Phase	In phase with original input	Out of phase with original input
Effect	Increases system gain	Decreases system gain
OPAM terminal	Output fed to non-inverting terminal	Output fed to inverting terminal
Stability	Can make system unstable	Stabilizes the system
Example	Oscillator circuits, microphone squealing	Thermostat, cruise control

3.7 Differential & Partial Differential Equations

Ordinary Differential Equation (ODE): Contains derivatives of dependent variable with respect to ONE independent variable.

$$M\ddot{x} + D\dot{x} + Kx = K_f(t) \text{ (Second-order ODE)}$$

Partial Differential Equation (PDE): Contains derivatives with respect to MORE THAN ONE independent variable.

$$\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0 \text{ (Laplace equation)}$$

Engineering Uses

- Most physical & chemical processes involve rate of change → requires differential equations
- Used to understand general effects of growth rate/trends
- Heat conduction, fluid dynamics, electromagnetic fields, structural mechanics

Past questions (10 of 10)

2019 F · 6 marks · repeated 5x

Explain feedback system with its application.

Answer

A **feedback system** is one where the output is sampled and fed back to the input to control system behavior.

Types

Positive Feedback: Output is in phase with input → increases system gain. Fed to non-inverting terminal of OPAM. Can make system unstable.

Negative Feedback: Output is out of phase with input → decreases system gain. Fed to inverting terminal. Stabilizes the system.

Application: Home Heating System

A thermostat-controlled furnace is a **negative feedback system**:

1. The furnace heats the room (output = room temperature)
2. Thermostat measures current temperature
3. If temperature < setpoint → furnace turns ON

4. If temperature > setpoint → furnace turns OFF
5. This creates a stable temperature around the setpoint

Application: Aircraft Autopilot

Gyroscope detects difference between actual and desired heading → sends signal to move control surfaces → airframe steers toward desired heading. This is a negative feedback loop that keeps the aircraft on course.

2014 F · 5 marks · repeated 3x

Explain continuous system simulation with suitable example.

Answer

Continuous System Simulation involves modeling systems where variables change continuously over time using differential equations.

Characteristics: Variables change smoothly, modeled by ODEs/PDEs, simulated using analog/digital/hybrid computers.

Example — Automobile Suspension System:

The suspension absorbs shock during vehicle movement on rough surfaces.

$$M\ddot{x} + D\dot{x} + Kx = Kf(t)$$

where M = mass, D = damping coefficient, K = spring constant, x = displacement, f(t) = road surface force.

Simulation using analog components:

1. Three signals ($M\ddot{x}$, $D\dot{x}$, Kx) are added using a **summer**
2. Output scaled by $1/M$ to produce $\ddot{x}$
3. Passed through **integrator** to get $\dot{x}$
4. Through **inverter** to get $-\dot{x}$
5. Through second **integrator** to get $-x$
6. Through **inverter** to get x

2014 F · 5 marks · repeated 3x

Why are differential equations so important in simulation? Explain with example.

Answer

Differential equations are crucial in simulation because:

1. **Dynamic Modeling:** They model how quantities change with respect to time/space (temperature changes, population growth, circuit behavior)
2. **Continuity:** Continuous systems like liquid flow and heat distribution involve quantities that change continuously — tracked by differential equations
3. **Predictive Power:** Once the DE is known, it can be solved numerically or analytically to predict future behavior from initial conditions
4. **Broad Application:** Used in engineering, physics, biology, economics — virtually every field

Example — Water Reservoir:

$$dh(t)/dt = (Q_{in}(t) - Q_{out}(t)) / A(h(t))$$

where $h(t)$ = water level at time t , Q_{in} = inlet rate, Q_{out} = outlet rate, $A(h)$ = cross-sectional area at height h . This ODE lets us simulate and predict water levels over time.

Example — Suspension System: $M\ddot{x} + D\dot{x} + Kx = K_f(t)$ describes the displacement of a vehicle over rough terrain, allowing engineers to optimize suspension parameters.

2025 M · 4 marks · repeated 2x

Describe the pure pursuit problem in continuous systems.

Answer

The **pure pursuit problem** is a classic continuous system simulation where a pursuer (e.g., fighter aircraft) constantly adjusts its direction to fly directly toward a moving target (e.g., bomber).

Setup: Fighter at $(0, 50)$, Bomber on x -axis. Fighter always points toward bomber's current position.

Assumptions: Same 2D plane, constant fighter speed, known target path, minimum firing distance = 10 units.

Formulas:

- Distance: $d = \sqrt{(x_b - x_f)^2 + (y_b - y_f)^2}$

- $\sin \theta = (y_b - y_f)/d$, $\cos \theta = (x_b - x_f)/d$
- $x_f(t+1) = x_f(t) + V_p \times \sin \theta$
- $y_f(t+1) = y_f(t) + V_p \times \cos \theta$

If $d \leq 100\text{m}$ → bomber shot down. If $d > 1000\text{m}$ → bomber escapes. If target path is straight → analytical solution. If curved → simulation needed.

2025 M · 8 marks

Compare analog vs. hybrid simulation methods, highlighting their use in solving differential equations.

Answer

Feature	Analog Simulation	Hybrid Simulation
Nature	Purely continuous signals	Combination of continuous + discrete
Components	Summer, Integrator, Inverter, Scale factor	Analog + Digital computer with DAC/ADC
Programmability	Non-programmable	Partially programmable (digital part)
Processing	Real-time continuous	Both real-time and stored-program
Accuracy	High accuracy, limited precision	Better precision (digital processing)
Differential Equations	Solved directly by physical components	Continuous part by analog, complex logic by digital
Example	$M\ddot{x} + D\dot{x} + Kx = K_f(t)$ — suspension system	Satellite simulation (equations of motion + digital control)
Limitation	Cannot handle digital/discrete logic	Requires high-speed DAC and ADC

For solving differential equations:

- **Analog:** Each term of the equation is represented by a physical component. Integrators solve $\int$, summers combine terms, scale factors apply constants. Output is a continuous voltage representing the solution.
- **Hybrid:** The continuous differential equation part is handled by analog components, while boundary conditions, logic decisions, and data logging are handled by the digital computer. This gives the real-time speed of analog with the flexibility of digital.

2015 F · 5 marks

Explain how continuous model can be solved numerically with an example

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 5 marks

Explain the analogy method of system study with suitable example

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 5 marks

Illustrate the possible event interactions in Hybrid Simulation

Practice question — model answer not written yet. Attempt it from the chapter notes.

2012 C · 5 marks

Solve equations by analog method: $\dot{X}_1 = -K_{11}X_1 + K_{12}X_2$; $\dot{X}_2 = K_{21}X_1 - (K_{22} + K_{23})X_2$; $\dot{X}_3 = K_{31}X_1$

Practice question — model answer not written yet. Attempt it from the chapter notes.

2010 F · 4 marks

Represent $Ax'' + Bx' + Cx + D = 0$ in analog computer block diagram.

Answer

Given: $Ax'' + Bx' + Cx + D = 0$

Rearranging: $Ax'' = -Bx' - Cx - D$

Therefore: $x'' = (-Bx' - Cx - D) / A$

Block Diagram Construction:

1. Start with three signals: $-Bx'$, $-Cx$, and $-D$
2. Feed all three into a **Summer (Adder)** → output = $-Bx' - Cx - D$

3. Pass through **Scale Factor (1/A)** → output = $x'' = (-Bx' - Cx - D)/A$

4. Pass x'' through **Integrator** → output = x'

5. Pass x' through **Inverter** → output = $-x'$

6. Pass $-x'$ through **Scale Factor (B)** → output = $-Bx'$ (feed back to summer)

7. Pass x' through second **Integrator** → output = x

8. Pass x through **Inverter** → output = $-x$

9. Pass $-x$ through **Scale Factor (C)** → output = $-Cx$ (feed back to summer)

10. Constant $-D$ is also fed to the summer

The feedback loop continuously solves the differential equation in real-time.

4.1 Elements of Queuing System

A **queuing system** (also known as a queuing model) is a mathematical concept used to analyze the behavior of waiting lines or queues.

Basic Structure

Customers → Arrival → Queue (Waiting Line) → Service Facility → Departure

Elements

1. **Calling Population:** The source of customers (finite or infinite)
2. **Arrival Process:** How customers arrive (Poisson, deterministic, general distribution)
3. **Queue:** The waiting line before service
4. **Service Mechanism:** Number of servers and how they operate
5. **Service Time Distribution:** Time to serve a customer (exponential, deterministic, normal)
6. **Queue Discipline:** Order of service (FIFO, LIFO, Random, Priority-based)
7. **System Capacity:** Maximum customers the system can hold
8. **Exit/Departure:** How customers leave after service

4.2 Characteristics of a Queuing System

The behaviour of a waiting line is fixed by a small set of characteristics. Every queuing problem is described by choosing one option from each row of the table below, and those choices are exactly what Kendall's notation has to record.

Characteristic	Options	Effect on the model
Arrival pattern	Poisson (random), deterministic (appointments), general	Poisson arrivals give exponential inter-arrival times
Service pattern	Exponential, deterministic, general	Exponential service means a few customers take much longer
Number of servers (channels)	Single or multiple	More servers cut waiting time but raise the cost
Number of service phases	Single phase or multiple phases	Multiple phases (registration → treatment) require a network of queues
Queue discipline	FIFO, LIFO, SIRO (random), priority	Priority changes the waiting time experienced by each class
System capacity	Finite (bounded) or infinite	Finite capacity causes blocking and balking
Calling population	Finite or infinite	With a finite population the arrival rate falls as the queue grows
Human behaviour	Balking, reneging, jockeying, collusion	Cannot be captured by the standard formulas
State of the system	Transient or steady state	Formulas such as $L = \rho/(1-\rho)$ hold only in steady state

Transient vs Steady State

The **transient state** is the early period after the system starts empty, when the statistics are still biased by that empty start (this is the initial-bias problem of section 7.4). The **steady state** is reached once the probability distribution of the system state stops changing with time. Analytic formulas are steady-state results, so a simulation must discard its warm-up period before the two can be compared fairly.

Human Behaviour in Queues

- **Balking:** the customer sees the long queue and refuses to join.
- **Reneging:** the customer joins but leaves before being served.

- **Jockeying:** the customer switches from one queue to another.
- **Collusion:** a group of customers joins the queue together (one person holds the place for several).

4.3 Types of Queuing System

Type	Description	Example
Single Channel, Single Server	One queue, one server	Single ATM machine
Multiple Channel, Single Server	Multiple queues, one server each	Bank with multiple tellers
Single Channel, Multiple Server	One queue, multiple servers	Airport check-in counters
Multiple Channel, Multiple Server	Multiple queues, multiple servers	Hospital emergency department

4.4 Kendall's Notation

A standard notation to describe characteristics of a queuing system:

A / S / C / K / N / D		
Symbol	Meaning	Possible Values
A	Arrival Process Distribution	M (Markovian/Poisson), D (Deterministic), G (General)
S	Service Time Distribution	M (Markovian/Exponential), D (Deterministic), G (General)
C	Number of Servers	1, 2, 3, ...
K	System Capacity	Optional, default = ∞
N	Population Size	Optional, default = ∞
D	Service Discipline	FIFO (default), LIFO, Priority

Examples
M/M/1: Poisson arrivals, Exponential service, 1 server, infinite capacity, infinite population, FIFO
M/M/8/20/500/LIFO: Poisson arrivals, Exponential service, 8 servers, capacity 20, population 500, LIFO
D/M/1/LIFO/20/95: Deterministic arrivals, Exponential service, 1 server, LIFO, capacity 20, population 95

4.5 Measurement of System Performance

Important Formulas (for M/M/1 queue)

These formulas are very important for exam problems!

Metric	Formula	Description
Arrival Rate (λ)	λ = customers/unit time	Average arrivals per unit time
Service Rate (μ)	μ = customers/unit time	Average customers served per unit time
Traffic Intensity (ρ)	$\rho = \lambda/\mu$	Must be < 1 for stable system
Avg customers in system (L)	$L = \rho/(1-\rho)$	Average number in entire system
Avg customers in queue (L_q)	$L_q = \rho^2/(1-\rho)$	Average waiting in queue only
Avg time in system (W)	$W = 1/\mu(1-\rho)$	Average total time spent
Avg waiting time in queue (W_q)	$W_q = \rho/\mu(1-\rho)$	Average wait before service
Idle probability (P_0)	$P_0 = 1-\rho$	Probability system is empty
n customers probability (P_n)	$P_n = (1-\rho)\rho^n$	Probability of n customers in system

Hospital emergency — M/M/1 performance measures worked · all seven measures from λ and μ

Arrivals **72 patients / day**

Service time **12 min / patient**

Servers **1 (M/M/1)**

1. $\lambda = 72 / (24 \times 60) = 72 / 1440 = 0.05$ patients per minute
 λ and μ must share ONE time unit — minutes here, because the service time is in minutes.
2. $\mu = 1 / 12 = 0.08333$ patients per minute
3. $\rho = \lambda / \mu = 0.05 / 0.08333 = 0.60$ $\rho < 1$, so the queue is stable — say this before quoting any other number.
4. $L = \rho/(1 - \rho) = 0.6/0.4 = 1.5$ patients in the system
5. $L_q = \rho^2/(1 - \rho) = 0.36/0.4 = 0.9$ patients waiting in the queue
6. $W_q = \rho/[\mu(1 - \rho)] = 0.6/0.03333 = 18$ minutes average wait
7. $W = 1/[\mu(1 - \rho)] = 1/0.03333 = 30$ minutes total time in the system
8. $P_0 = 1 - \rho = 0.40 \rightarrow$ the single server is idle 40% of the time
Check: $W = W_q + 1/\mu = 18 + 12 = 30$ ✓ and $L = \lambda W = 0.05 \times 30 = 1.5$ ✓

$\rho = 0.6$ $L = 1.5$ · $L_q = 0.9$ · $W_q = 18$ min · $W = 30$ min · $P_0 = 0.40$

4.6 Application of Queuing System

- **Telecommunications:** Optimizing network traffic, data packet handling
- **Healthcare:** Patient flow, reducing waiting times
- **Manufacturing:** Assembly line optimization, reducing bottlenecks
- **Transportation:** Bus stations, traffic lights, airport security
- **Banking:** Customer service, teller allocation

4.7 Markov Chain

A **Markov chain** is a mathematical model for a process that moves step by step through various states.

Properties

1. The outcome of each experiment is one of a set of **discrete states**
2. The outcome depends only on the **present state**, not any past state (memoryless property)
3. The transition probabilities remain **constant** from one transition to next

Steps for Future Prediction

1. Create current state distribution matrix Q
2. Create the transition probability matrix P
3. Calculate $Q = Q \times P$ (probability vector after n transitions)

Markov chain — Honda / Yamaha purchase after 3 purchases 2026 pattern · P^3 worked by hand

Honda → Honda **0.70**

Honda → Yamaha **0.30**

Yamaha → Yamaha **0.80**

Yamaha → Honda **0.20**

Current state $Q = [1, 0]$ (Honda user)

1. Transition matrix $H \ Y \ H \ 0.70 \ 0.30 \ P = \begin{bmatrix} 0.70 & 0.30 \\ 0.20 & 0.80 \end{bmatrix}$ Each ROW must add to 1: it is the distribution of the next purchase given the current one.
2. $P^2 = P \times P$ row H: $[0.70 \times 0.70 + 0.30 \times 0.20, 0.70 \times 0.30 + 0.30 \times 0.80] = [0.55, 0.45]$ row Y: $[0.20 \times 0.70 + 0.80 \times 0.20, 0.20 \times 0.30 + 0.80 \times 0.80] = [0.30, 0.70]$
3. $P^3 = P^2 \times P$ row H: $[0.55 \times 0.70 + 0.45 \times 0.20, 0.55 \times 0.30 + 0.45 \times 0.80] = [0.475, 0.525]$
4. $Q = Q \times P^3 = [1, 0] \times P^3 =$ first row of $P^3 = [0.475, 0.525]$ Starting as a Honda user you only need row H — the second row answers the same question for a Yamaha user.

$P(\text{Yamaha after 3 purchases}) = 0.525$ **52.5%**

Past questions (9 of 9)

2019 F · 5 marks · repeated 4x

Markov Chain (Short Note)

Answer

A **Markov chain** is a mathematical model for a process that moves step by step through various states. It consists of **states** and **transition probabilities**.

Properties:

1. Outcome of each experiment is one of a set of **discrete states**
2. Outcome depends only on the **present state**, not any past state (memoryless/Markov property)
3. Transition probabilities remain **constant** from one transition to next

Steps for prediction:

1. Create initial state vector Q
2. Create transition probability matrix P
3. Calculate $Q = Q \times P^n$ for n transitions

Applications: Weather forecasting, traffic systems, telephone systems, PageRank algorithm, stock market analysis.

2014 F · 10 marks · repeated 4x

Computing facility of PU has large no. of personal computers. 50% of days: 0 failures, 30%: 1 failure, 20%: 2 failures. Repair: 40% in 1 day, 35% in 2 days, 25% in 3 days. Verify service person's claim of being overloaded.

Answer

Failure arrival rate (λ):

$$\lambda = 0.50 \times 0 + 0.30 \times 1 + 0.20 \times 2 = \mathbf{0.70 \text{ machines per day}}$$

Average service time (T):

$$T = 1 \times 0.40 + 2 \times 0.35 + 3 \times 0.25 = 0.40 + 0.70 + 0.75 = \mathbf{1.85 \text{ days per machine}}$$

Service rate (μ):

$$\mu = 1/T = 1/1.85 = \mathbf{0.54 \text{ machines per day}}$$

System utilization (ρ):

$$\rho = \lambda/\mu = 0.70/0.54 = \mathbf{1.30}$$

Since $\rho > 1$, the system is OVERLOADED!

The arrival rate exceeds the service rate. The queue will grow without bound.

Average machines waiting:

$$L_q = \lambda^2 / \mu(\mu - \lambda) = 0.70^2 / (0.54 \times (0.54 - 0.70)) = 0.49 / (0.54 \times (-0.16)) = 0.49 / (-0.0864) = \mathbf{-5.67}$$

Negative sign indicates system is overloaded and unstable

Average waiting time:

$$W_q = \lambda / \mu(\mu - \lambda) = 0.70 / (0.54 \times (0.54 - 0.70)) = 0.70 / (-0.0864) = \mathbf{-8.10 \text{ days}}$$

Conclusion: Since $\rho = 1.30 > 1$, the service person's claim is CORRECT. The system is overloaded — a second service person must be hired.

2015 F, 2011 F, 2010 F · 5 marks · repeated 3x

Explain the arrival and departure routine queuing model for a banking system

Practice question — model answer not written yet. Attempt it from the chapter notes.

2025 M · 4 marks · repeated 2x

What are the elements of a queuing system?

Answer

The key elements of a queuing system are:

1. **Calling Population:** Source of customers that arrive (finite or infinite)
2. **Arrival Process:** Pattern of arrivals (Poisson, deterministic, general)
3. **Queue:** The waiting line before service begins
4. **Service Mechanism:** Number of servers and how they operate
5. **Service Time Distribution:** Statistical distribution of service duration (exponential, deterministic, normal)
6. **Queue Discipline:** Rule for selecting next customer (FIFO, LIFO, Priority, Random)
7. **System Capacity:** Maximum customers the system can hold
8. **Exit/Departure:** How customers leave after being served

2019 F · 5 marks · repeated 2x

Kendall's Notation (Short Note)

Answer

Kendall's Notation is a standard notation used to describe queuing systems in the form: **A/S/C/K/N/D**

- **A** = Arrival process distribution (M=Markovian/Poisson, D=Deterministic, G=General)
- **S** = Service time distribution (M=Exponential, D=Deterministic, G=General)
- **C** = Number of servers
- **K** = System capacity (default: ∞)
- **N** = Population size (default: ∞)
- **D** = Service discipline (default: FIFO)

Examples:

- M/M/1 — Simplest queuing model (Poisson arrival, Exponential service, 1 server)
- M/M/8/20/500/LIFO — 8 servers, capacity 20, population 500, LIFO discipline

2012 C · 10 marks · repeated 2x

For 8 customers, the inter-arrival time and service time in a single counter of a bank are given. Develop a simulation table. Inter-arrival: 0.4, 1.2, 0.5, 1.7, 0.2, 1.6, 0.2, 1.8. Service: 2.0, 0.7, 0.2, 1.1, 3.7, 0.6, 0.4, 0.7

Answer

Formulas:

- Actual Arrival Time = Cumulative of inter-arrival time
- Service Start Time = $\max(\text{Arrival Time}, \text{Previous Service End Time})$
- Service End Time = Service Start Time + Service Time
- Waiting Time = Service Start Time - Actual Arrival Time

Cust	Inter-arrival	Arrival	Service Start	Service Time	Service End	Wait
1	0.4	0.4	0.4	2.0	2.4	0
2	1.2	1.6	2.4	0.7	3.1	0.8
3	0.5	2.1	3.1	0.2	3.3	1.0
4	1.7	3.8	3.8	1.1	4.9	0
5	0.2	4.0	4.9	3.7	8.6	0.9
6	1.6	5.6	8.6	0.6	9.2	3.0
7	0.2	5.8	9.2	0.4	9.6	3.4
8	1.8	7.6	9.6	0.7	10.3	2.0

i) Average waiting time = $(0+0.8+1.0+0+0.9+3.0+3.4+2.0)/8 = 11.1/8 = 1.3875$ min

ii) Probability of waiting = $6/8 = 0.75 = 75\%$ (6 out of 8 customers waited)

iii) Average customers in queue = Total waiting time / Total arrival span = $11.1 / (7.6-0.4) = 11.1/7.2 = 1.542$

2025 M · 8 marks

Simulate a bank queuing system using Kendall's notation. Discuss performance metrics like average wait time. [4+4]

Answer

Part 1: Bank Queuing System in Kendall's Notation

A typical bank with single teller: **M/M/1**

- **M** (first): Customers arrive following Poisson distribution
- **M** (second): Service times follow Exponential distribution
- **1**: Single server (one teller)
- Infinite capacity, infinite population, FIFO discipline (defaults)

If the bank has 3 tellers with capacity 50: **M/M/3/50**

Part 2: Performance Metrics

For M/M/1 with $\lambda=0.05$, $\mu=0.0833$ (hospital example):

Metric	Formula	Value
Traffic Intensity (ρ)	$\lambda/\mu = 0.05/0.0833$	0.6
Avg customers in system (L)	$\rho/(1-\rho)$	1.5
Avg customers in queue (L_q)	$\rho^2/(1-\rho)$	0.9
Avg time in system (W)	$1/\mu(1-\rho)$	30 min
Avg wait time in queue (W_q)	$\rho/\mu(1-\rho)$	18 min
Idle probability (P_0)	$1-\rho$	0.4 (40%)

2011 C · 2+3 marks

What is queuing system? Why do we use queuing system in simulation?

Practice question — model answer not written yet. Attempt it from the chapter notes.

2010 C · 10 marks

Library queuing simulation for 15 customers — calculate average waiting time, average time in system, probability of server being busy

Practice question — model answer not written yet. Attempt it from the chapter notes.

5.1 Model Building

Model building is the process of creating a simulation model that accurately represents the real-world system.

Steps of Model Building

1. **Observe the real system:** Study interactions among components, collect data on their behavior. But observation alone seldom yields sufficient understanding.
2. **Question individuals:** Operators, technicians, repair/maintenance personnel, engineers, supervisors, and managers — collect their experience and domain knowledge.

Construct a conceptual model: Based on assumptions about components and structure. Identify:

- Assumptions of system components
- Structure assumptions (interactions between components)
- Input parameters and data assumptions

4. **Construct an operational model:** Use simulation software or programming language. Incorporate the conceptual assumptions into the tool being used.

Model Building Diagram

Three-Level Architecture

Real System → (Conceptual Validation) → **Conceptual Model** → (Model Verification) → **Operational Model**

And: **Operational Model** → (Calibration & Validation) → **Real System**

5.2 Verification & Validation

Key Distinction (Very Important for Exams!)

Verification: "Are we building the model *correctly*?" (Correct I/P → Correct O/P)

Validation: "Are we building the *correct* model?" (Does model match real world?)

Feature	Verification	Validation
Question	Is the model built correctly?	Is the correct model built?
Compares	Conceptual model vs Operational model	Operational model vs Real system
Focus	Code correctness, logic errors	Real-world accuracy, predictive power
Method	Testing, debugging, code review	Calibration, comparison with real data

5.3 Verification of Simulation Models

Verification tests whether the model is correctly implemented in simulation software. Key considerations:

1. The operational model should be checked by **someone other than the developer**, preferably an expert in simulation software
2. Make a **flow diagram** of all logically possible actions the system can perform when an event occurs
3. Closely **examine the output** for a variety of input sets
4. Have the operational model **print input parameters** when simulation ends
5. Make the operational model as **self-documenting** as possible
6. Verify that **animated operational model** imitates the actual system
7. Use a **debugger** during simulation model building
8. **Graphical interfaces** are recommended for visualization

Three Classes of Verification Techniques

1. **Finding deficiencies in the model design** — walk through the conceptual model and the code with someone who knows the system, looking for logic that cannot occur in reality.
2. **Model testing with known inputs** — run the model with inputs whose correct output can be worked out by hand, then compare. This is the single most convincing verification evidence.
3. **Determining whether the model is internally consistent** — no contradiction between two parts of the code, and the module interfaces match the conceptual model.

Static vs Dynamic Verification

Static (no execution)	Dynamic (model is run)
Structured walkthroughs of the conceptual model	Debugging with a debugger / trace facility
Code review by an expert other than the author	Printing input parameters and intermediate state at run time
Flow diagrams of every logically possible event action	Comparing output for a variety of input sets
Checking that the code is self-documenting	Animated operational model vs the real system

Additional Checks: Degenerate, Extreme-Condition and Continuity Tests

Degenerate Tests

Feed the model input values that make its behaviour analytically known and see whether the output matches. Example: set the arrival rate equal to zero in a queuing model — the queue length must stay at zero for the whole run. Example: make every service time identical — the output must stop being random.

Extreme-Condition Tests

Push the model into the extreme ranges of the input variables to confirm that it still behaves reasonably, and that it fails gracefully outside the range for which it was built. Example: an arrival rate so high that the queue grows without bound — the model should show the queue growing steadily, not produce nonsense or crash.

Continuity Test

Change an input variable slightly and confirm the output also changes slightly. A large jump in the output for a tiny change in the input is a symptom of a coding error. Example: raising the arrival rate from 4.9 to 5.0 per hour must not change the average waiting time from 10 minutes to 500 minutes.

Trace and Operational Graphics

A **trace** is a time-stamped log of the state, the event list and the statistics as each event executes; reading it exposes errors that no amount of output inspection would reveal. **Operational graphics** (animation, dynamic status displays) let an observer watch the modelled system behave and spot behaviour that is structurally wrong.

Note: some textbooks group degenerate, extreme-condition and continuity tests under validation rather than verification. The tests themselves are identical — only the label differs, so answer with the definitions above and say which grouping you are using.

5.4 Calibration & Validation of Models

Calibration

Calibration is the process of comparing the model with the real-world system and **adjusting parameters** so that the model output matches real-world data. It's an iterative process of parameter tuning.

Validation Process (Naylor & Finger's 3 Steps)

Three Steps for Model Validation

1. **High Face Validity:** The model should appear reasonable to users and domain experts. Involve potential users in model construction for realism.

Validate Model Assumptions:

- *Structural Assumptions:* Components and how the system operates (e.g., one queue vs many, FCFS vs priority)
- *Data Assumptions:* Statistical analysis of input/output parameters (e.g., inter-arrival times, service times)

3. **Validate I/O Transformation:** Compare model output with real system output. The model should make accurate predictions for the range of input data sets of interest.

Sensitivity Analysis for Validation

Check model validity by testing if it responds logically to changes:

Example

In a queuing system: if arrival rate increases → the queue length and waiting time should also increase. If the model doesn't show this behavior, it may not be valid.

Iterative Validation Process

Validation is iterative — compare, revise, repeat:

1. **Initial Model** → compare to reality → **Revise**
2. **First Revised Model** → compare to reality → **Revise**
3. **Second Revised Model** → compare to reality → Continue until satisfactory

Each revision improves the model's accuracy until it adequately represents the real system for the intended purpose.

Three Ways to Validate the Input–Output Transformation

1. **Prediction of the real system (strongest).** Use the model to predict the performance of the system under a new set of conditions, then run the real system and compare. Predictive validation is the most convincing evidence of validity.
2. **Historical data validation.** Drive the model with input data recorded from the real system and compare the model's output with the output the real system actually produced for the same period.
3. **The Turing test.** Ask people who know the system well to discriminate between the model's output and the real system's output. If the expert cannot reliably tell them apart, the model is said to possess sufficient validity for the purpose.

The Turing Test in Simulation

Two forms are used:

1. **Interactive Turing test** — an expert interacts with the model (or its animation) without knowing which system is behind the interface.
2. **Retrospective Turing test** — the expert is given printed output of the model and of the real system, in random order, and asked to identify which is which.

The test is subjective, so it is used together with objective statistical comparisons, never alone.

Objective vs Subjective Validation

Subjective	Objective
Face validity by experts and users	Statistical comparison of model output with system output
Turing test	Confidence intervals, hypothesis tests, K–S and chi-square comparisons
Review of the assumptions document	Time-series methods — cross-correlation, spectral analysis, Theil's inequality coefficient

Statistical Methods for Comparing Model and System Output

1. **Graphical comparison** — plot the model output and the system output (or their trend lines) on the same axes; a close match over time is supporting evidence.
2. **Confidence intervals** — build a confidence interval for the difference between the model mean and the system mean; if it contains zero, the difference is not significant.
3. **Hypothesis tests for means** — the two-sample t test compares the mean of the model output with the mean of the real-system output.
4. **Hypothesis tests for distributions** — the two-sample chi-square test or the two-sample Kolmogorov–Smirnov test compares the whole distribution of model output with that of the system (see the K–S test in Chapter 6).

5. **Change detection** — two independent samples of model output (different random numbers) are checked to ensure that the model itself is not drifting.

Sargent's Model Confidence Levels

Level	Meaning
0	The model is not validated — clearly inappropriate or based on unverified assumptions.
1	Validation has been performed by people knowledgeable about the system (face validity) — useful but weak.
2	Extensive validation of the assumptions and of the input-output transformation has been carried out.
3	The model gives acceptable results for a wide variety of test conditions and is accepted for the intended purpose. This is the target of a good simulation study.

Why Validation is Difficult — Sources of Invalidity

- **Erroneous assumptions** about the structure of the system or about the input distributions.
- **Lack of knowledge** about the real system's behaviour, so the model is built on guesswork.
- **Inappropriate level of model detail** — too coarse to capture the behaviour that matters, or too detailed to be estimated reliably.
- **Poor data collection** — the recorded data may itself be biased or measured incorrectly.
- **Misinterpretation of the results** — the model may be valid but the conclusions drawn from it are not.
- **Cost and time** — full validation is expensive, so a model is usually validated only for the specific purpose it was built for.

Putting It Together

Verification vs Calibration vs Validation

Term	Question answered	Compared
Conceptual validation	Is the model theory sound?	Real system ↔ Conceptual model
Verification	Was the model built correctly?	Conceptual model ↔ Operational model
Calibration	Which parameter values make the model match reality?	Operational model ↔ Real system (parameter tuning)
Validation	Was the correct model built?	Operational model ↔ Real system (overall accuracy)

All four run iteratively until the model is accepted at confidence level 3 for its intended purpose, and the assumptions and validation evidence are then written up in the assumptions document and the validation document.

Past questions (5 of 5)

2025 M · 4 marks

Differentiate between verification and validation of simulation models.

Answer

Feature	Verification	Validation
Definition	"Building the model correctly"	"Building the correct model"
Question	Does the code correctly implement the conceptual model?	Does the model accurately represent the real system?
Compares	Conceptual model vs Operational (computerized) model	Operational model vs Real-world system
Focus	Internal correctness — logic errors, bugs, coding mistakes	External accuracy — real-world fidelity, predictive power
Methods	Debugging, code review, testing with known inputs, flow diagrams, animation	Calibration, comparison with real data, sensitivity analysis, face validity
Example	Checking if arrival process code correctly generates Poisson distributed arrivals	Comparing simulated average wait times with actual measured wait times in the real bank

Key relationship: Verification ensures correctness of implementation. Validation ensures correctness of the model itself. Both are necessary — a verified but invalid model is useless, and an unverified valid concept is buggy.

General · 10 marks

Explain model building, verification, and validation with diagram.

Answer

Model Building Steps:

- Observe** the real system, collect data

2. **Question** domain experts (operators, engineers, managers)
3. **Construct conceptual model** with assumptions (component, structure, data)
4. **Construct operational model** in simulation software

Diagram:

Real System $\longleftrightarrow$ Conceptual Model $\longleftrightarrow$ Operational Model

- **Conceptual Validation:** Real System $\leftrightarrow$ Conceptual Model
- **Model Verification:** Conceptual Model $\leftrightarrow$ Operational Model
- **Calibration & Validation:** Operational Model $\leftrightarrow$ Real System

Verification = Building the model correctly (code implements the concept)

Validation = Building the correct model (model matches reality)

Naylor & Finger's 3 Steps:

1. Build model with high face validity (reasonable to experts)
2. Validate model assumptions (structural + data)
3. Validate I/O transformation (compare outputs with real system)

Iterative Process: Initial Model $\rightarrow$ Compare $\rightarrow$ Revise $\rightarrow$ First Revised Model $\rightarrow$ Compare $\rightarrow$ Revise $\rightarrow$ Continue until satisfactory.

General · 5 marks

Explain the various verification techniques used for a simulation model.

Answer

Verification asks "Was the model built correctly?" — does the operational (computerised) model faithfully implement the conceptual model? The standard techniques are:

1. Checks on the model design

- Have the operational model checked by **someone other than the developer**, preferably an expert in simulation software.

- Draw a **flow diagram** of every logically possible action the system can perform when each event occurs.
- Make the operational model as **self-documenting** as possible.

2. Model testing with known inputs

- Run the model with inputs whose correct output can be computed by hand, then compare.
- Examine the **output for a variety of input sets** and confirm it is sensible.
- Have the model **print input parameters** at the end of the run, so the run can be reproduced.

3. Internal consistency

- Use a **debugger** while building and running the model.
- Use **graphical interfaces / animation** so the behaviour can be watched.

4. Special verification tests

Technique	What it does
Degenerate tests	Feed inputs whose result is analytically known (zero arrival rate → empty queue).
Extreme-condition tests	Push inputs to their limits and confirm the model still behaves reasonably.
Continuity test	A tiny change in an input must produce a tiny change in the output — a large jump indicates a coding error.
Trace / operational graphics	Log the state and event list as each event executes, or animate the model, to expose structural errors.

Note: some textbooks group the degenerate, extreme-condition and continuity tests under validation. State which grouping you are using and give the same definitions.

General · 7 marks

Explain the approaches used for validating a simulation model. What is the Turing test?

Answer

Validation asks "Was the correct model built?" — does the model, as a whole, represent the real system accurately enough for its intended purpose?

1. Naylor & Finger's three-step approach

1. **Build with high face validity** — involve users and domain experts so the model appears reasonable to people who know the system.
2. **Validate the model assumptions** — structural assumptions (one queue vs many, FCFS vs priority) and data assumptions (statistical analysis of inter-arrival and service times).
3. **Validate the input–output transformation** — compare model output with real-system output over the range of input data of interest.

2. Three ways to validate the input–output transformation

- **Predictive validation (strongest):** use the model to predict the system's performance under new conditions, then run the real system and compare.
- **Historical data validation:** drive the model with recorded real input data and compare its output with what the real system actually produced.
- **Turing test:** ask knowledge people to distinguish model output from real-system output.

3. The Turing test

People who are knowledgeable about the system are asked to discriminate between the output of the model and that of the real system. If they cannot tell them apart reliably, the model possesses sufficient validity for the purpose. Two forms exist: an **interactive** Turing test (the expert interacts with the model or its animation) and a **retrospective** Turing test (the expert is shown printed output of both systems in random order). The test is subjective, so it is always used alongside objective statistical comparisons.

4. Objective statistical comparison methods

- **Graphical comparison** of model and system output, or of their trend lines.
- **Confidence intervals** for the difference between model and system means — if the interval contains zero the difference is not significant.
- **Two-sample t test** to compare means, and the **two-sample chi-square or Kolmogorov–Smirnov test** to compare whole distributions.
- **Time-series methods** — cross-correlation, spectral analysis, Theil's inequality coefficient.

5. Sensitivity analysis

Check that the model responds logically to changes, e.g. in a queuing system a higher arrival rate must increase the queue length and waiting time. If it does not, the model may be invalid.

6. Model confidence levels (Sargent)

Level 0 not validated, level 1 validation by people knowledgeable about the system, level 2 extensive validation of assumptions and I/O transformation, level 3 acceptable results for a wide variety of test conditions. A good study targets level 3.

General · 5 marks

Why is validation of a simulation model difficult? Discuss the sources that can make a model invalid.

Answer

Validation is difficult because there is no single test that proves a model correct, the real system may not be available for experimentation, and cost and time limit how much evidence can be collected. Common sources of invalidity are:

1. **Erroneous assumptions** — wrong structure, or wrong input distributions (for example assuming exponential service times when they are not).
2. **Lack of knowledge of the real system** — the model is built on guesswork because the system's behaviour is not well understood.
3. **Inappropriate level of model detail** — too coarse to capture the behaviour that matters, or so detailed that its parameters cannot be estimated reliably.
4. **Poor data and data collection** — recorded data may be biased, incomplete, or measured with the wrong instrument; garbage in, garbage out.
5. **Incorrectly modelled distributions** — the fitted input distribution is rejected by goodness-of-fit tests but used anyway.
6. **Misinterpretation of the results** — the model may be valid while the conclusions drawn from it are not.
7. **Cost and time** — exhaustive validation is expensive, so a model is normally validated only for the specific purpose it was built for.

Mitigation: use several independent validation approaches together (face validity, validated assumptions, historical data, predictive validation, Turing tests, statistical comparison), keep an assumptions document and a validation document, and state the confidence level at which the model is accepted.

6.1 Random Numbers

Random numbers are numbers that occur in a sequence such that two conditions are met:

1. **Uniformity:** Each number is uniformly distributed on $(0,1)$ — every number has equal probability
2. **Independence:** Each number is statistically independent of all others — no detectable pattern

Highest Weightage Chapter (12 Marks)

This is the most important chapter! LCM generation and statistical tests are asked in nearly every exam.

6.2 Random Number Tables

Pre-computed tables of random digits. Numbers are selected from these tables for simulation. However, tables are limited and not practical for large simulations.

6.3 Pseudo Random Numbers

Pseudo-random numbers are generated by a deterministic algorithm but appear random. They are called "pseudo" because they are not truly random — they are produced by a mathematical formula.

Properties

- **Fast generation:** Computed quickly by formula
- **Reproducible:** Same seed produces same sequence (useful for debugging)
- **Long period:** Sequence repeats after many numbers
- **Pass statistical tests:** Appear random by all standard tests

6.4 Generation of Random Numbers

Linear Congruential Method (LCM) — Most Important!

$$X_{n+1} = (a \times X_n + c) \bmod m$$

where:

- X_0 = seed (starting value)
- a = multiplier
- c = increment
- m = modulus

Mixed Congruential: $c \neq 0$ (has increment)

Multiplicative Congruential: $c = 0$ (no increment)

LCM — the numbers the 2025 model question gives 2025 M · 8 marks

Seed X_0 27

Multiplier a 17

Increment c 43

Modulus m 100

1. $X_1 = (17 \times 27 + 43) \bmod 100 = 502 \bmod 100 = 2$ $R_1 = X_1 / m = 2 / 100 = 0.02$ $17 \times 27 = 459$; $459 + 43 = 502$; $502 - 5 \times 100 = 2$.
2. $X_2 = (17 \times 2 + 43) \bmod 100 = 77 \bmod 100 = 77$ $R_2 = 77 / 100 = 0.77$ The product is already below m , so the modulus changes nothing.
3. $X_3 = (17 \times 77 + 43) \bmod 100 = 1352 \bmod 100 = 52$ $R_3 = 52 / 100 = 0.52$ $1352 - 13 \times 100 = 52$.
4. $X_4 = (17 \times 52 + 43) \bmod 100 = 927 \bmod 100 = 27$ $R_4 = 27 / 100 = 0.27$ $927 - 9 \times 100 = 27$ — this is the seed X_0 again.
5. $X_5 = (17 \times 27 + 43) \bmod 100 = 502 \bmod 100 = 2$ $R_5 = 2 / 100 = 0.02$ The sequence has closed on itself.

Five random numbers 2, 77, 52, 27, 2 normalized 0.02, 0.77, 0.52, 0.27, 0.02

The sequence returns to $X_0 = 27$ after only four steps, so the **period is 4**. It therefore cannot supply more than a handful of values before it starts repeating — which is why these parameters are a demonstration, not a usable generator.

LCM — the numbers worked in class Lesson example

Seed X_0 30

Multiplier a 12

Increment b 21

Modulus m 100

1. $X_1 = (12 \times 30 + 21) \bmod 100 = 381 \bmod 100 = 81$ $R_1 = 81 / 100 = 0.81$
2. $X_2 = (12 \times 81 + 21) \bmod 100 = 993 \bmod 100 = 93$ $R_2 = 0.93$ $993 - 9 \times 100 = 93$.
3. $X_3 = (12 \times 93 + 21) \bmod 100 = 1137 \bmod 100 = 37$ $R_3 = 0.37$
4. $X_4 = (12 \times 37 + 21) \bmod 100 = 465 \bmod 100 = 65$ $R_4 = 0.65$

$$5. X_{100} = (12 \times 65 + 21) \bmod 100 = 801 \bmod 100 = 1 \quad R_{100} = 0.01$$

Five random numbers **81, 93, 37, 65, 1** normalized **0.81, 0.93, 0.37, 0.65, 0.01**

Multiplicative Congruential Generator — finding the Period

With $c = 0$ the generator is $X_{i+1} = (a \times X_i) \bmod m$. Because $b = 0$ the seed can never be produced again by accident, so the period has to be found by generating until **the seed itself reappears** — that is the point at which the whole sequence must repeat. The question below is the one that has been asked the most often (2010 C and 2011 F).

Period of a multiplicative CGM — two seeds 2010 C · 2011 F

Multiplier a **13**

Increment b **0**

Modulus m **2⁶ = 64**

Seeds X_0 **2 and 4**

- $X_{i+1} = (a \times X_i) \bmod m = (13 \times X_i) \bmod 64$ $b = 0$, so this is the pure multiplicative form.
- $X_1 = 2$
 $X_2 = 13 \times 2 = 26 \bmod 64 = 26$
 $X_3 = 13 \times 26 = 338 \bmod 64 = 18$
 $X_4 = 13 \times 18 = 234 \bmod 64 = 42$
 $X_5 = 13 \times 42 = 546 \bmod 64 = 34$
 $X_6 = 13 \times 34 = 442 \bmod 64 = 58$
 $X_7 = 13 \times 58 = 754 \bmod 64 = 50$
 $X_8 = 13 \times 50 = 650 \bmod 64 = 10$
 $X_9 = 13 \times 10 = 130 \bmod 64 = 2$

Subtract the largest multiple of 64 that fits: $338 - 5 \times 64 = 18$; $234 - 3 \times 64 = 42$; $546 - 8 \times 64 = 34$; $442 - 6 \times 64 = 58$; $754 - 11 \times 64 = 50$; $650 - 10 \times 64 = 10$; $130 - 2 \times 64 = 2$.
- $X_1 = 2 = X_9 \rightarrow$ the seed has reappeared at step 8 period = 8
- $X_1 = 4$
 $X_2 = 13 \times 4 = 52 \bmod 64 = 52$
 $X_3 = 13 \times 52 = 676 \bmod 64 = 36$
 $X_4 = 13 \times 36 = 468 \bmod 64 = 20$
 $X_5 = 13 \times 20 = 260 \bmod 64 = 4$

$676 - 10 \times 64 = 36$; $468 - 7 \times 64 = 20$; $260 - 4 \times 64 = 4$.
- $X_1 = 4 = X_5 \rightarrow$ the seed has reappeared at step 4 period = 4

Period $X_0 = 2 \rightarrow 8$ $X_0 = 4 \rightarrow 4$

Read the Result Carefully

The period is the number of steps needed to come back to the seed, *counting the return itself*. With $X_0 = 2$ the generator produces 26, 18, 42, 34, 58, 50, 10 and then 2 again — so eight steps, and the whole history repeats from there. A different seed gives a different period: **the period belongs to the seed, not just to the parameters**.

Does the Generator have Full Period?

A generator has **full period** only if it can produce every value from 0 to $m - 1$ before repeating. For the mixed form there is a standard test — **the Hull–Dobell conditions**:

c must be coprime to m · $a - 1$ must be divisible by every prime factor of m · if $4 \mid m$ then $4 \mid (a - 1)$

Generate with full period? Check the conditions Lesson example

Multiplier a **12**

Increment c **5**

Modulus m **26**

Seeds X₀ 6 and 7

1. $m = 26 = 2 \times 13 \rightarrow$ prime factors 2 and 13 $a - 1 = 11$ 11 is not divisible by 2, and not by 13 The second Hull–Dobell condition already fails, so full period is impossible. Generating the numbers confirms it.
2. $X_0 = 6$ $X_1 = (12 \times 6 + 5) \bmod 26 = 77 \bmod 26 = 25$ $R_1 = 25/26 = 0.96$ $X_2 = (12 \times 25 + 5) \bmod 26 = 305 \bmod 26 = 19$ $R_2 = 19/26 = 0.73$ $X_3 = (12 \times 19 + 5) \bmod 26 = 233 \bmod 26 = 25$ $R_3 = 0.9677 - 2 \times 26 = 25$; $305 - 11 \times 26 = 19$; $233 - 8 \times 26 = 25$. $X_4 = X_1$, so the sequence is trapped in the two-value cycle $25 \leftrightarrow 19$ — it never even returns to the seed.
3. $X_0 = 7$ $X_1 = (12 \times 7 + 5) \bmod 26 = 89 \bmod 26 = 11$ $R_1 = 11/26 = 0.42$ $X_2 = (12 \times 11 + 5) \bmod 26 = 137 \bmod 26 = 7$ $R_2 = 7/26 = 0.27137 - 5 \times 26 = 7 = X_0$, the seed again.

Verdict No full period — period 2 for both seeds

Only the numbers 25, 19 (or 7, 11) are ever produced, instead of all 26 values. Round to two decimals for R, or keep the exact fraction — **state which convention you are using**, because 25/26 is 0.96, not 0.95.

Combined Linear Congruential Method

One generator with a short period is a weak generator. The **combined** (or composite) method runs two generators side by side and adds their outputs, so the combined sequence only repeats when *both* do — this is the trick behind most modern generators.

$$X_{n+1} = (a_1 X_n + b_1) \bmod m_1 \quad Y_{n+1} = (a_2 Y_n + b_2) \bmod m_2 \quad Z_n = (X_n + Y_n) \bmod m, \quad m = \max(m_1, m_2)$$

Combined LCM — combined output Z_n Lesson example

a_1, b_1, m_1 5, 1, 16

X_0 1

a_2, b_2, m_2 3, 2, 16

Y_0 1

$m = \max(m_1, m_2)$ 16

1. $n = 0$: $Z_0 = (X_0 + Y_0) \bmod 16 = (1 + 1) \bmod 16 = 2$ $R_0 = Z_0 / m = 2 / 16 = 0.125$
2. $X_1 = (5 \times 1 + 1) \bmod 16 = 6$ $Y_1 = (3 \times 1 + 2) \bmod 16 = 5$ $Z_1 = (X_1 + Y_1) \bmod 16 = (6 + 5) \bmod 16 = 11$ $R_1 = 11 / 16 = 0.6875$ Advance both generators first, then combine.
3. $X_2 = (5 \times 6 + 1) \bmod 16 = 31 \bmod 16 = 15$ $Y_2 = (3 \times 5 + 2) \bmod 16 = 17 \bmod 16 = 1$ $Z_2 = (X_2 + Y_2) \bmod 16 = (15 + 1) \bmod 16 = 0$ $R_2 = 0 / 16 = 0.0016 \bmod 16 = 0$ — a legitimate output, not an error. Do not drop it.
4. $X_3 = (5 \times 15 + 1) \bmod 16 = 76 \bmod 16 = 12$ $Y_3 = (3 \times 1 + 2) \bmod 16 = 5$ $Z_3 = (12 + 5) \bmod 16 = 1$ $R_3 = 1 / 16 = 0.0625$

Combined numbers 2, 11, 0, 1 normalized 0.125, 0.6875, 0.00, 0.0625

6.5 Mid-Square Random Number Generator

Proposed by **Von Neumann** in 1946. One of the earliest methods.

Algorithm

1. Start with an n-digit seed
2. Square the seed
3. Extract the middle n digits → this is the next random number
4. Repeat

Mid-square — 4-digit seed $X_{\blacksquare} = 1234$ Lesson example

Seed $X_{\blacksquare} \mathbf{1234}$

Digits kept $\mathbf{4}$

Number of values $\mathbf{5}$

1. $X_{\blacksquare} = (1234)^2 = 1\,522\,756$ pad to 8 digits → 01522756 middle 4 digits → 5227 A 4-digit number squared gives at most 8 digits, so always write it as exactly 8 — pad with a leading zero when the square is short. The middle 4 of 01522756 are the digits in positions 3–6: 5 2 2 7.
2. $X_{\blacksquare} = (5227)^2 = 27\,321\,529$ middle 4 digits → 3215
3. $X_{\blacksquare} = (3215)^2 = 10\,336\,225$ middle 4 digits → 3362
4. $X_{\blacksquare} = (3362)^2 = 11\,303\,044$ middle 4 digits → 3030
5. $X_{\blacksquare} = (3030)^2 = 9\,180\,900$ pad to 8 digits → 09180900 middle 4 digits → 1809 3030^2 is only 7 digits, so the leading zero is what makes the middle-4 extraction correct. Without it you would take 1809 out of the wrong position.

Random numbers **5227, 3215, 3362, 3030, 1809**

Problems with Mid-Square

Can degenerate to 0 (once 0000 appears, all subsequent values are 0). Short period. Not recommended for serious simulation.

6.6 Qualities of an Efficient Random Number Generator

1. **Uniformity:** Numbers should be uniformly distributed
2. **Independence:** No correlation between successive numbers
3. **Long period:** Should not repeat for a long sequence
4. **Reproducibility:** Same seed → same sequence (for debugging)
5. **Fast computation:** Should be computationally efficient
6. **Non-repeating:** For the length of use needed

6.7–6.8 Testing for Randomness

Pseudo-random numbers are only useful if they survive statistical testing. Every test sets up a null hypothesis and either rejects it or fails to reject it:

The Hypothesis Testing Framework

H₀: the numbers are uniformly distributed and independent (they *are* random).

H₁: the numbers are not uniformly distributed, or not independent.

Decision rule: compute the test statistic. If it is **less than** the critical value, **fail to reject H₀** — the generator passes. If it **exceeds** the critical value, **reject H₀** — the generator fails.

A stated confidence level of 95% means $\alpha = 0.05$, so there is a 5% chance of rejecting a good generator (Type I error).

Categories of Tests

- **Uniformity tests** — is the distribution flat over $[0,1]$? → Frequency test, Chi-Square, Kolmogorov–Smirnov
- **Independence tests** — is each number unrelated to the ones before it? → Auto-Correlation, Run Test, Gap Test
- **Pattern tests** — do the digit patterns match theory? → Poker Test

No single test proves randomness. A generator is accepted only after it passes the whole battery of tests, and the tests are complementary: chi-square tells you about the distribution but says nothing about the order in which numbers appear.

Frequency (Uniformity) Test

The simplest uniformity check: divide $[0,1]$ into k equal intervals, count how many numbers fall in each, and confirm every interval holds roughly N/k values. The chi-square test in section 6.9 is the formal version of this idea.

6.9 Chi-Square Test

$$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$$

where O_{ij} = observed frequency, E_{ij} = expected frequency

Steps

1. Divide the range [0, 99] into n equal intervals
2. Count observed frequency (O_{ij}) in each interval
3. Calculate expected frequency: $E_{ij} = N/n$ (total numbers / intervals)
4. Calculate χ^2 statistic
5. Compare with $\chi^2(\alpha, n-1)$ from table
6. If calculated $\chi^2 < \text{table value} \rightarrow$ numbers are random (fail to reject H_0)

Chi-square test on 50 two-digit random numbers 10 intervals · $\alpha = 0.05$

N50

Intervals **n10 (width 10)**

$E_{ij} = N/n = 50/10 = 5$

df = n – 1 = 9

The 50 numbers, in the order given:

36, 91, 51, 02, 54, 06, 58, 06, 58, 02, 54, 01, 48, 97, 43, 22, 83, 25, 79, 95,
42, 87, 73, 17, 02, 42, 95, 38, 79, 29, 65, 09, 55, 97, 39, 83, 31, 77, 17, 62,
03, 49, 90, 37, 13, 17, 58, 11, 51, 92

1. H_0 : the numbers are uniformly distributed on [0, 1) H_1 : they are not uniformly distributed $\alpha = 0.05$ for a 95% confidence level (level of significance = 1 – level of confidence).

Count how many numbers fall in each interval — tally the list, do not estimate.

Interval	O_{ij}	E_{ij}	$O_{ij} - E_{ij}$	$(O_{ij} - E_{ij})^2$	$(O_{ij} - E_{ij})^2 / E_{ij}$
00 – 09	8	5	+3	9	1.80
10 – 19	5	5	0	0	0.00
20 – 29	3	5	–2	4	0.80
30 – 39	5	5	0	0	0.00
40 – 49	5	5	0	0	0.00
50 – 59	8	5	+3	9	1.80
60 – 69	2	5	–3	9	1.80

70 – 79	4	5	–1	1	0.20
80 – 89	3	5	–2	4	0.80
90 – 99	7	5	+2	4	0.80
Total	50	50	—	—	8.00

2. ΣO_{ij} must equal N. The ledger above has $8 + 5 + 3 + 5 + 5 + 8 + 2 + 4 + 3 + 7 = 50$ — always check this before trusting the statistic, because one mis-tallied number changes the answer.
3. $\chi^2 = \Sigma (O_{ij} - E_{ij})^2 / E_{ij} = 1.80 + 0.00 + 0.80 + 0.00 + 0.00 + 1.80 + 1.80 + 0.20 + 0.80 + 0.80$ $\chi^2 = 8.00$
4. Rejection region: $\chi^2(\alpha, n - 1) = \chi^2(0.05, 9) = 16.919$ Critical values are always read with $df = n - 1 = 10 - 1 = 9$, not 10.

$\chi^2 = 8.00 < 16.919$ **Fail to reject H_0 — the numbers are uniform**

Why the number of intervals matters

The same 50 values grouped into 5 intervals give $O_{ij} = 13, 8, 13, 6, 10$ against $E_{ij} = 10$, so $\chi^2 = 3.80$ with $df = 4$ and a critical value of 9.488 — the same conclusion by a different route. **State the grouping you used**, because χ^2 and its degrees of freedom are not comparable between two different groupings.

6.10 Testing for Auto-Correlation

The **auto-correlation test** asks whether a number is correlated with the number / places after it. The uniformity tests look at the values; this one looks at the **order**, so it catches a generator that produces a beautifully flat but predictable sequence.

Procedure

1. Choose the position i and the lag m (the question usually supplies them, e.g. "the 3rd, 8th, 13th ... numbers", which means $i = 3$ and $m = 5$)
2. Find the largest M with $i + (M + 1)m \leq N$
3. Form the products $R_i R_{i+m}$, $R_{i+m} R_{i+2m}$, ... up to $R_{i+Mm} R_{i+(M+1)m}$ — there are $M + 1$ of them
4. Compute ρ , then σ_ρ , then Z
5. Compare $|Z|$ with $Z_{\{\alpha/2\}}$

$$\rho = \left[\frac{1}{M+1} \cdot \sum (R_{i+km} \times R_{i+(k+1)m}) \right] - 0.25 \quad \sigma_\rho = \sqrt{\frac{(13M + 7)}{(12(M + 1))}} \quad Z = \rho / \sigma_\rho \quad \text{reject } H_0 \text{ if } |Z| > Z_{\{\alpha/2\}}$$

The constant **0.25** is the value the sum would have for perfectly independent numbers (the mean of $R_i R_{i+m}$ is $\frac{1}{4}$), so ρ measures the deviation from independence. At $\alpha = 0.05$, $Z_{\{0.025\}} = 1.96$.

Auto-correlation at lag 5, starting at the 3rd number

N30

Start position $i=3$

Lag $m=5$

$Z_{\{\alpha/2\}}=1.96$

The 30 numbers, read left to right as $R_1 \dots R_{30}$:

R_1-R_6 0.12 0.01 0.23 0.28 0.89 R_6-R_{11} 0.31 0.64 0.28 0.83 0.93 $R_{11}-R_{16}$ 0.99
 0.15 0.33 0.35 0.91 $R_{16}-R_{21}$ 0.41 0.60 0.27 0.75 0.38 $R_{21}-R_{26}$ 0.68 0.49 0.05
 0.43 0.95 $R_{26}-R_{31}$ 0.50 0.19 0.36 0.69 0.87

1. H_0 : the 3rd, 8th, 13th ... numbers are not correlated H_1 : they are correlated
2. $i + (M + 1)m \leq N$ $3 + (M + 1) \times 5 \leq 30$ $(M + 1) \times 5 \leq 27$ $M + 1 = 5$, so $M = 427/5 = 5.4$, and $M + 1$ must be a whole number, so take 5 — that is the number of products to compute.
3. The indices are $i + km$ for $k = 0 \dots M + 1$: 3, 8, 13, 18, 23, 28 ($28 \leq 30$, so the choice of M is sound) $R_3 = 0.23$ $R_8 = 0.28$ $R_{13} = 0.33$ $R_{18} = 0.27$ $R_{23} = 0.05$ $R_{28} = 0.36$
4. $R_3 R_8 = 0.23 \times 0.28 = 0.0644$ $R_8 R_{13} = 0.28 \times 0.33 = 0.0924$ $R_{13} R_{18} = 0.33 \times 0.27 = 0.0891$ $R_{18} R_{23} = 0.27 \times 0.05 = 0.0135$ $R_{23} R_{28} = 0.05 \times 0.36 = 0.0180$ $\Sigma = 0.2774$ Five products, because $M + 1 = 5$.
5. $\rho = 0.2774 / 5 - 0.25 = 0.05548 - 0.25 = -0.1945$
6. $\sigma_\rho = \sqrt{[(13 \times 4 + 7) / (12 \times 5)]} = \sqrt{(59/60)} = \sqrt{0.9833} = 0.9916$
7. $Z = \rho / \sigma_\rho = -0.1945 / 0.9916 = -0.196$
8. $|Z| = 0.196 \leq 1.96 \rightarrow$ fail to reject H_0

$Z_5 = -0.196$ the numbers are independent — no auto-correlation at lag 5

Do not use m where M belongs

Two different symbols are in play and swapping them is the usual mistake: **m is the lag** (5 here) and **M is the number of steps along the lag minus one** (4 here). σ_{ρ_m} uses M , and the divisor of ρ_m is $M + 1$, which is also the count of products. If your divisor and your product count disagree, one of them is wrong.

6.11 Poker Test

Classifies n-digit numbers by their digit patterns and tests using chi-square.

For 3-digit numbers, classifications:

- **All different** (e.g., 123): $P = 10 \times 9 \times 8 / 10^3 = 0.720$
- **One pair** (e.g., 112): $P = 0.270$
- **Three of a kind** (e.g., 111): $P = 0.010$

Compare observed vs expected using χ^2 test.

Poker test on 1000 three-digit numbers

N 1000

Digits per number 3

Categories k 3

df = k - 1 = 2

1. H_0 : the digit patterns follow the theoretical probabilities (the numbers are random) H_a : the digit patterns do not
2. Expected probability of each category, out of $10^3 = 1000$ possible numbers: All different $\rightarrow 10 \times 9 \times 8 = 720 \rightarrow P = 720/1000 = 0.720$ One pair $\rightarrow 10 \times 3 \times 9 = 270 \rightarrow P = 270/1000 = 0.270$ Three of a kind $\rightarrow 10 = 10 \rightarrow P = 10/1000 = 0.010$

Category	Observed (O)	Expected (E)
All different	680	720
One pair	289	270
Three of a kind	31	10

 total = 1000/1000 = 1.000 Read "one pair" as: choose the repeated digit (10), choose which of the 3 positions holds the odd digit (3), choose that odd digit (9) — which is why 000 and 111 fall under "three of a kind", not here.
3. Observed counts, converted to the expected frequencies $E_i = P_i \times N$: Category O E $O - E$
 $(O - E)^2 / E$ All different 680 720 -40 1600 2.222 One pair 289 270 +19 361 1.337 Three of a kind 31 10 +21 441 44.100

Category	O	E	$(O-E)^2/E$
All different	680	720	2.222
One pair	289	270	1.337
Three of a kind	31	10	44.100

 $\chi^2 = 47.659$ $\sum O = 1000 = N$ ✓ and $\sum E = 1000$ ✓, so the table is complete.
4. Rejection region: $\chi^2(\alpha, k - 1) = \chi^2(0.05, 2) = 5.991$ df is the number of categories minus one = 3 - 1 = 2. It is not the number of digits.

$\chi^2 = 47.659 > 5.991$ **Reject H_0 — the digit patterns are not random**

Look at which category caused it: the "three of a kind" count of 31 is more than three times the expected 10. One badly-weighted category is enough to fail the test even though the other two are close to expectation.

6.12 Run Test (Runs Up and Down Test)

The **run test** checks independence using only the *order* of the numbers, never their values. A **run** is a maximal succession of numbers moving in the same direction.

Procedure

1. Write the sequence $R_1, R_2, \dots, R_N$
2. Mark each adjacent difference as + (the number rose) or – (it fell)
3. Count the total number of runs a — each maximal block of identical signs is one run
4. Compute the expected number of runs and its standard deviation
5. Compute Z and compare it with $Z_{\{\alpha/2\}}$

$$\mu_a = (2N - 1)/3 \quad \sigma^2_a = (16N - 29)/90 \quad Z = (a - \mu_a)/\sigma_a$$

Decision: if $-Z_{\{\alpha/2\}} \leq Z \leq Z_{\{\alpha/2\}}$ the numbers are independent. At $\alpha = 0.05$, $Z_{\{\alpha/2\}} = 1.96$; at $\alpha = 0.02$, $Z_{\{\alpha/2\}} = 2.33$.

Run test — a 10-number sequence $\alpha = 0.05$, $Z_{\{\alpha/2\}} = 1.96$

N10

Signs counted **N – 1 = 9**

$\alpha 0.05$

$Z_{\{\alpha/2\}} 1.96$

0.42, 0.18, 0.65, 0.03, 0.79, 0.55, 0.32, 0.11, 0.88, 0.61

1. H_0 : the random numbers are independent H_1 : they are not independent
2. Write + where the number rose and – where it fell: 0.42→0.18 – 0.18→0.65 + 0.65→0.03 – 0.03→0.79 + 0.79→0.55 – 0.55→0.32 – 0.32→0.11 – 0.11→0.88 + 0.88→0.61 – sign string: – + – + – – – + – Nine comparisons for ten numbers. Watch the two equal-and-adjacent case here: 0.18 → 0.65 rises, and 0.55 → 0.32 falls.
3. Group the signs into maximal blocks of the same sign: [–] [+] [–] [+] [– – –] [+] [–] number of runs $a = 7$
4. $\mu_a = (2N - 1)/3 = (2 \times 10 - 1)/3 = 19/3 = 6.333$ $\sigma^2_a = (16N - 29)/90 = (160 - 29)/90 = 131/90 = 1.456$ $\sigma_a = \sqrt{1.456} = 1.207$
5. $Z = (a - \mu_a)/\sigma_a = (7 - 6.333)/1.207 = 0.667/1.207 = 0.55$
6. $-1.96 \leq 0.55 \leq 1.96 \rightarrow$ fail to reject H_0

$Z = 0.55$ the sequence shows no evidence of dependence

Run test — the 40-number sequence from class $N = 40 \cdot \alpha = 0.05$

N40

Signs counted **N – 1 = 39**

$Z_{\{\alpha/2\}} 1.96$

0.41 0.68 0.89 0.94 0.74 0.41 0.55 0.62 0.36 0.27 0.19 0.72 0.75 0.08 0.54 0.02
0.01 0.36 0.16 0.28 0.18 0.01 0.95 0.61 0.18 0.47 0.23 0.32 0.82 0.53 0.31 0.12

0.73 0.04 0.83 0.45 0.13 0.57 0.63 0.29

1. H_0 : the random numbers are independent H_1 : they are not independent
2. Mark every move, then group the signs: + + + - - + + - - + + - + - - + - - + - - + - + + - - - + - + - -
+ + - runs: +++ | ---++ | ---++ | -+ | ---+ | -+ | ---+ | ---+ | -+ | ---+ | +- number of runs $a = 24$ With $N = 40$
there are 39 signs. Count in blocks of similar signs rather than one by one — a single missed sign changes a .
3. $\mu_a = (2 \times 40 - 1)/3 = 79/3 = 26.33$ $\sigma^2_a = (16 \times 40 - 29)/90 = 611/90 = 6.789$ $\sigma_a = \sqrt{6.789} = 2.61$
4. $Z = (a - \mu_a)/\sigma_a = (24 - 26.33)/2.61 = -2.33/2.61 = -0.89$
5. $-1.96 \leq -0.89 \leq 1.96 \rightarrow$ fail to reject H_0

$Z = -0.89$ the numbers are independent

Watch the Counting Convention

Decide up front whether the first or last observation starts a run, then apply the convention consistently. Different textbooks count one run more or fewer; the worked answers here count every maximal block of like signs, including single-element blocks.

6.13 Kolmogorov–Smirnov (K–S) Test

The K–S test compares the **empirical cumulative distribution** $F_n(x)$ with the **theoretical cumulative distribution** $F(x)$. For numbers on $[0,1)$, $F(x) = x$. Unlike chi-square it needs **no grouping into intervals**, which makes it the stronger test — and it can be applied to any hypothesised distribution, not just the uniform.

Procedure

1. Sort the N numbers in ascending order: $R_1 \leq R_2 \leq \dots \leq R_N$
2. For each i , the empirical CDF *after* R_i is i/N and *before* R_i is $(i - 1)/N$; the theoretical value is $F(R_i) = R_i$
3. Compute both deviations for every i and keep the largest absolute one
4. Compare D with the critical value
5. If $D \leq$ the critical value, fail to reject H_0 — the numbers are uniformly distributed

$$D = \max_i \{ |i/N - R_i|, |R_i - (i - 1)/N| \} \quad D_{\text{critical}} = D_\alpha / \sqrt{N}$$

For large samples $D_{0.05} = 1.36$ and $D_{0.01} = 1.63$; small samples ($N < 35$) use the exact table, which gives about 0.41 for $N = 10$ at $\alpha = 0.05$.

K–S test — one-sample, $N = 102011$ C · 2010 C

N10

$\alpha 0.05$

$D_\alpha = 1.36$ for large N

0.24, 0.89, 0.11, 0.61, 0.23, 0.86, 0.41, 0.64, 0.50, 0.65

1. H_0 : the numbers are uniformly distributed on $[0, 1)$ H_1 : they are not uniformly distributed
2. Sort the numbers — they must be ascending before anything is tabulated: 0.11, 0.23, 0.24, 0.41, 0.50, 0.61, 0.64, 0.65, 0.86, 0.89

i	R_i	i/N	$F(R_i) = R_i$	$ i/N - R_i $	$(i-1)/N$	$ R_i - (i-1)/N $
1	0.11	0.1	0.11	0.01	0.0	0.11
2	0.23	0.2	0.23	0.03	0.1	0.13
3	0.24	0.3	0.24	0.06	0.2	0.04
4	0.41	0.4	0.41	0.01	0.3	0.11
5	0.50	0.5	0.50	0.00	0.4	0.10
6	0.61	0.6	0.61	0.01	0.5	0.11
7	0.64	0.7	0.64	0.06	0.6	0.04

8	0.65	0.8	0.65	0.15	0.7	0.05
9	0.86	0.9	0.86	0.04	0.8	0.06
10	0.89	1.0	0.89	0.11	0.9	0.01

3. $F(R_i) = R_i$ because the theoretical distribution is uniform on $[0, 1]$. Both deviations are taken at every i : the empirical CDF can sit above or below the theoretical line, and D is the worse of the two.
4. $D_{\text{upper}} = \max(i/N - R_i) = 0.15$ (at $i = 8$) $D_{\text{lower}} = \max(R_i - (i-1)/N) = 0.13$ (at $i = 2$) $D = \max(D_{\text{upper}}, D_{\text{lower}}) = 0.15$
5. Critical value for large samples: $D_{\text{critical}} = D_{\alpha} / \sqrt{N} = 1.36 / \sqrt{10} = 1.36 / 3.162 = 0.430$ (small-sample exact table for $N = 10 \approx 0.41$)

$D = 0.15 < 0.430$ **fail to reject H_0 — consistent with a uniform $[0,1]$ generator**

K–S test — the short sequence worked in class $D_{\alpha} = 0.565$ given

$N=6$

$\alpha=0.05$

$D_{\alpha}=0.565$ (given)

0.63, 0.49, 0.24, 0.57, 0.71, 0.89

1. H_0 : the numbers are uniformly distributed on $[0, 1]$ H_1 : they are not uniformly distributed
2. Sorted: 0.24, 0.49, 0.57, 0.63, 0.71, 0.89
3. $i \quad R_i \quad i/N \quad i/N - R_i \quad R_i - (i-1)/N$
1 0.24 0.167 -0.073 0.24
2 0.49 0.333 -0.163 0.32
3 0.57 0.500 -0.070 0.24
4 0.63 0.667 0.037 0.13
5 0.71 0.833 0.123 0.043
6 0.89 1.000 0.110 0.057
 i/N is 1/6, 2/6, 3/6 ... not 0.2, 0.4 — with $N = 6$ the thirds have to be exact enough to compare with R_i .
4. $D_{\text{upper}} = \max(i/N - R_i) = 0.123$ $D_{\text{lower}} = \max(R_i - (i-1)/N) = 0.32$ $D = \max(0.123, 0.32) = 0.32$
5. Test: $D < D_{\alpha} \rightarrow 0.32 < 0.565 \rightarrow$ do not reject H_0

$D = 0.32 < 0.565$ **the random numbers are uniform**

Two K–S forms, one idea

The **one-sample** test above compares the numbers with a claimed distribution; the **two-sample** test compares two data sets against each other, using $D_{\alpha} = C(\alpha) \times \sqrt{[(n_1 + n_2)/(n_1 n_2)]}$ with $C(0.05) = 1.36$ — for $n_1 = n_2 = 124$ that gives $D_{\alpha} = 0.172$. Both use the same statistic and the same decision rule.

Choosing the Right Test

Test	What it verifies	Statistic	Critical value
Frequency / Chi-Square	Uniformity across grouped intervals	$\chi^2 = \sum (O_i - E_i)^2 / E_i$	$\chi^2(\alpha, k - 1)$
Kolmogorov–Smirnov	Uniformity of the whole CDF, no grouping	$D = \max F_n - F $	$D_{\alpha} / \sqrt{N}$ (1.36 at $\alpha = 0.05$)

Auto-Correlation	Independence of numbers separated by lag l	$Z_l = \rho_l / \sigma_{\rho_l}$	$Z_{\{\alpha/2\}}$ (1.96)
Run Test	Independence — order of up and down moves	$Z = (a - \mu_a) / \sigma_a$	$Z_{\{\alpha/2\}}$ (1.96)
Poker Test	Independence of digit patterns inside each number	$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$	$\chi^2(\alpha, k - 1)$
Gap Test	Length of gaps between numbers in a chosen range	$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$	$\chi^2(\alpha, k - 1)$

Exam Tip

LCM generation, the chi-square test and the auto-correlation test are the most frequently asked numericals from this chapter. The **run test** and the **Kolmogorov–Smirnov test** are the two items students most often skip, and they have both already appeared (2015 F and 2011 C / 2010 C) — do not leave them out.

Past questions (27 of 27)

2014 F · 5 marks · repeated 3x

What are the qualities of good random number generators? What are the methods used for testing?

Answer

Part 1 — Qualities of a good random number generator

1. **Uniformity:** values are uniformly distributed over (0, 1).
2. **Independence:** no correlation between successive values.
3. **Long period:** the sequence does not repeat within the length of any run.
4. **Reproducibility:** the same seed regenerates the same sequence, which is essential for debugging and for controlled experiments.
5. **Fast and cheap generation:** millions of values may be required.
6. **Small memory requirement** and **portability** across machines.
7. **Ability to provide several independent streams** for the different random components of the model.

Part 2 — Methods used for testing

Test	What it verifies
Frequency test	That values are spread evenly over (0, 1)
Chi-square test	Uniformity, by comparing observed and expected frequencies in k intervals
Kolmogorov–Smirnov test	Uniformity of the whole cumulative distribution, with no grouping
Auto-correlation test	Independence of values separated by a lag l
Run test (runs up and down)	Independence, from the pattern of rises and falls in the sequence
Gap test	Independence, from the length of gaps between values in a chosen range
Poker test	Independence of the digit patterns inside each number

A generator is accepted only if it passes the whole battery, since each test is sensitive to a different kind of defect.

2014 F · 5 marks · repeated 3x

Test whether the 2nd, 7th, 12th and so on numbers in the sequence are auto correlated. ($\alpha=0.05$). [30 numbers given]

Answer

The **auto-correlation test** checks whether numbers separated by a fixed lag are correlated. When the question asks about the 2nd, 7th, 12th ... numbers, the lag is $l = 5$ and only those numbers form the comparison series.

Steps

- Pick the numbers at positions 2, 7, 12, 17, 22, 27 from the 30-value sequence.
- Form the M pairs of consecutive values from this series: (2, 7), (7, 12), (12, 17), (17, 22), (22, 27). Here $M = 5$.
- Compute the estimated auto-correlation

$$\rho_{\text{AC}} = (1/M) \sum_{i=1}^M (R_i \times R_{i+1}) - 0.25$$

- Compute the standard deviation of ρ_{AC} and the test statistic:

$$\sigma_{\rho} = \sqrt{(13M + 7) / (12(M + 1))} \quad Z = \rho / \sigma_{\rho}$$

With $M = 5$: $\sigma_{\rho} = \sqrt{(13 \times 5 + 7) / (12 \times 6)} = \sqrt{72/72} = 8.485/72 = \mathbf{0.1178}$.

5. Compare $|Z|$ with $Z_{\{\alpha/2\}} = Z_{\{0.025\}} = \mathbf{1.96}$ for a two-tailed test at $\alpha = 0.05$.

Decision

- If $|Z| \leq 1.96$, the values are **not auto-correlated** and the numbers are accepted as independent.
- If $|Z| > 1.96$, the numbers are auto-correlated and the generator fails the test of independence.

The calculation of ρ itself uses only the five products obtained in step 2 — multiply each pair, sum the products, divide by M and subtract 0.25. In this test the statistic is extremely sensitive to the ordering of the values, which is exactly what the chi-square test cannot see.

2011 C · 4+1 marks · repeated 3x

Explain the Mid square random number generator with example. What is the problem with this generator?

Answer

Mid-square random number generator (proposed by von Neumann, 1946):

1. Start with an n -digit seed.
2. Square the seed to obtain a $2n$ -digit number.
3. Extract the middle n digits — this is the next random number.
4. Repeat, using the extracted value as the new seed. Divide by 10^n to normalize.

Example with a 4-digit seed:

Step	Seed	Squared	Middle 4 digits
1	7182	51581124	5811
2	5811	33767721	7677
3	7677	58936329	9363

Problem with the generator: the sequence can **degenerate**. If the middle digits ever become 0000 the square is 00000000 and every subsequent value stays 0, so the generator stops producing random numbers. Other weaknesses are the very short period for many seeds, sensitivity to the choice of seed, and no guarantee of uniformity. Because of this it has been replaced by the linear congruential method.

2015 F, 2011 F · 2+2 marks · repeated 2x

What are the properties of random numbers? Why are random numbers generated by a computer called Pseudo-random numbers?

Answer

Properties of random numbers (Banks):

1. **Uniformity:** the numbers must be uniformly distributed on the interval $(0, 1)$ — every value is equally likely, so no sub-interval is favoured.
2. **Independence:** each number must be statistically independent of every other number — there must be no detectable pattern or correlation between successive values.

These two properties are exactly what the statistical tests of section 6.7 onwards check.

Why computer-generated numbers are called pseudo-random:

Because they are produced by a **deterministic mathematical formula** (LCM, mid-square, etc.) from a starting seed. Given the same seed and the same formula, the identical sequence is reproduced every time. Such numbers therefore only *appear* random and pass the statistical tests for randomness — they are not truly random. The prefix *pseudo* (false, apparent) states precisely this: deterministic origin, random appearance.

True randomness requires a physical source (radioactive decay, thermal noise). Its practical disadvantages are that the sequence cannot be reproduced for debugging, and it cannot be generated fast enough for a large simulation.

2014 F, 2012 C · 5 marks · repeated 2x

Short Note: Poker test

Answer

Poker test — a test of independence for random numbers which, instead of looking at the magnitude of the numbers, classifies them by the **pattern of their digits** and compares the observed pattern frequencies with the theoretical probabilities using the chi-square test. It takes its name from the hands of a poker game.

Method

1. Group the random digits into numbers of d digits (commonly 3 or 5).
2. Define the pattern classes for those numbers. For 3-digit numbers there are three classes: all digits different, exactly one pair of like digits, and all three digits alike.
3. Compute the theoretical probability of each class. For 3-digit numbers:
 - All different: $10 \times 9 \times 8/10^3 = 0.720$
 - Exactly one pair: 0.270
 - Three alike: $10/10^3 = 0.010$
4. Count the observed frequency O_i of each class in the sample.
5. Compute the expected frequencies $E_i = P_i \times N$, where N is the number of numbers.
6. Compute $\chi^2 = \sum (O_i - E_i)^2/E_i$ with $k - 1$ degrees of freedom.
7. Compare with the critical value and decide.

Example

For 1000 three-digit numbers with observed counts 680, 289 and 31 against expected 720, 270 and 10:

$$\chi^2 = 1600/720 + 361/270 + 441/10 = 2.222 + 1.337 + 44.100 = \mathbf{47.659}, \text{ compared with } \chi^2(0.05, 2) = 5.991.$$

Since $47.659 > 5.991$ the numbers are **not independent** — the excess of numbers with three like digits is the cause.

Significance and limitation

The test is valuable because it detects dependence that the frequency tests cannot see: a generator can produce an evenly spread set of values whose digits are nevertheless patterned. Its limitation is that the result depends on the number of digits chosen per group, and it needs a large sample so that every expected frequency is at least 5.

2012 C · 5+5 marks · repeated 2x

Define Pseudo-random numbers. Explain the qualities of an efficient random number generator

Answer

Pseudo-random numbers are numbers generated by a deterministic algorithm that appear to be random — uniformly distributed on (0, 1) and independent of each other — but are completely reproducible from the seed.

Qualities of an efficient random number generator:

1. **Uniformity:** the numbers must be uniformly distributed over (0, 1).
2. **Independence:** no correlation between successive numbers.
3. **Long period:** the sequence must not repeat within the length of any run that will be made.
4. **Reproducibility:** the same seed must regenerate the same sequence, so runs can be repeated and debugged.
5. **Fast generation:** each number must cost very little computation, because a simulation may need millions of them.
6. **Low memory:** the generator should keep only a small amount of state between calls.
7. **Passes statistical tests:** it must satisfy the uniformity and independence tests (chi-square, K–S, auto-correlation, poker, run test).
8. **Portability:** it must work identically on different machines and compilers, independent of word length.
9. **Multiple independent streams:** the ability to produce several disjoint streams from one generator, needed for the different random components of a model.

The **Lehmer (Linear Congruential) generator** and the **combined LCG (L'Ecuyer)** are the classical designs that satisfy these requirements; the mid-square method satisfies almost none of them.

2025 M · 8 marks

Generate 5 pseudo-random numbers using the Linear Congruential Method with $X_0=27$, $a=17$, $c=43$, $m=100$.

Answer

Formula: $X_{i+1} = (a \times X_i + c) \bmod m = (17 \times X_i + 43) \bmod 100$

n	X_i	$a \times X_i + c$	$X_{i+1} = \text{result} \bmod 100$	$R_i = X_{i+1}/m$
0	27	$17 \times 27 + 43 =$ $459 + 43 = 502$	2	0.02

1	2	$17 \times 2 + 43 = 34 + 43 = 77$	77	0.77
2	77	$17 \times 77 + 43 = 1309 + 43 = 1352$	52	0.52
3	52	$17 \times 52 + 43 = 884 + 43 = 927$	27	0.27
4	27	$17 \times 27 + 43 = 459 + 43 = 502$	2	0.02

5 pseudo-random numbers: 2, 77, 52, 27, 2

Normalized [0,1): 0.02, 0.77, 0.52, 0.27, 0.02

Note: The sequence has period 4 (repeats after $27 \rightarrow 2 \rightarrow 77 \rightarrow 52 \rightarrow 27$). This is a short period because the parameters don't satisfy full-period conditions.

2019 F · 5 marks

Use LCM to generate random numbers. Seed $X_0 = 6, 7$; $a = 12$, $c = 5$, $m = 26$.

Answer

Formula: $X_{n+1} = (12 \times X_n + 5) \bmod 26$

For $X_0 = 6$:

n	X_n	$12 \times X_n + 5$	$X_{n+1} \bmod 26$
0	6	77	$77 \bmod 26 = 25$
1	25	305	$305 \bmod 26 = 19$
2	19	233	$233 \bmod 26 = 25$

Sequence: 25, 19, 25, 19, ... (period = 2)

For $X_0 = 7$:

n	X_n	$12 \times X_n + 5$	$X_{n+1} \bmod 26$
0	7	89	$89 \bmod 26 = 11$
1	11	137	$137 \bmod 26 = 7$

Sequence: 11, 7, 11, 7, ... (period = 2)

2019 F · 7 marks

Use Chi-Square test to test 50 random numbers at 95% confidence. Numbers:

36,91,51,02,54,06,58,06,58,02,54,01,48,97,43,22,83,25,79,95,42,87,73,17,02,42,95,38,79,29,65,09,55,97,39,83,31,77

Answer

H₀: Numbers are uniformly distributed (random)

H₁: Numbers are not random

Step 1: Divide [00-99] into 5 equal intervals

Step 2: Count observed frequencies

Interval	Numbers	O _{ij}	E _{ij} (50/5)	(O _{ij} -E _{ij}) ² /E _{ij}
00-19	02,06,06,02,01,17,02,09,13,03,13,17,11	13	10	0.90
20-39	36,22,25,29,38,39,31,37 8	8	10	0.40
40-59	51,54,58,58,54,48,43,42,42,55,49,58,51	12	10	0.90
60-79	79,73,65,77,62,79	6	10	1.60
80-99	91,97,83,95,87,83,95,97,90,92	10	10	0.00

$$\chi^2 = 0.90 + 0.40 + 0.90 + 1.60 + 0.00 = 3.80$$

Step 3: At $\alpha=0.05$, $df=5-1=4$: $\chi^2(0.05, 4) = 9.488$

Step 4: Since $3.80 < 9.488 \rightarrow$ Fail to reject H₀

Conclusion: The numbers ARE random at 95% confidence level.

2015 F · 5 marks

Based on the run test, check whether a given sequence of 40 numbers is random or not. [$Z_{(\alpha/2)} = 1.96$]

Answer

Purpose: the run test examines the *order* of the numbers rather than their values, so it detects dependence that chi-square cannot see.

Formula:

$$\mu_a = (2N - 1)/3 \quad \sigma^2_a = (16N - 29)/90 \quad Z_{\blacksquare} = (a - \mu_a)/\sigma_a$$

where **a** = total number of runs (maximal blocks of consecutive + and – signs) and N = number of observations.

Steps

1. Compare each number with the next one and write + if it increases, – if it decreases.
2. Count the runs: every maximal block of identical signs counts as one run.
3. Compute μ_a and σ_a from N.
4. Compute $Z_{\blacksquare} = (a - \mu_a)/\sigma_a$.
5. Compare with $Z_{\{\alpha/2\}} = 1.96$ (two-tailed, $\alpha = 0.05$).

Illustration of the counting step

For the 10-number sequence 0.42, 0.18, 0.65, 0.03, 0.79, 0.55, 0.32, 0.11, 0.88, 0.61 the signs are
– + – + – – – + –
which group into [–] [+] [–] [+] [– – –] [+] [–] → **a = 7**.

With N = 10: $\mu_a = 19/3 = 6.333$, $\sigma^2_a = 131/90 = 1.456$, $\sigma_a = 1.207$, so
 $Z_{\blacksquare} = (7 - 6.333)/1.207 = \mathbf{0.55}$. Since $|0.55| < 1.96$ we **fail to reject $H_{\blacksquare}$** — no evidence of dependence.

Applying it to the given 40 numbers

Carry out steps 1 and 2 on all 40 values to obtain a (the number of runs). With N = 40:

- $\mu_a = (2 \times 40 - 1)/3 = 79/3 = 26.333$
- $\sigma^2_a = (16 \times 40 - 29)/90 = 611/90 = 6.789$, so $\sigma_a = 2.606$
- $Z_{\blacksquare} = (a - 26.333)/2.606$

Conclusion rule: if $-1.96 \leq Z_{\blacksquare} \leq 1.96$ the sequence is accepted as random (independent); otherwise reject $H_{\blacksquare}$ and declare the numbers non-random. For example, $a = 24$ gives $Z_{\blacksquare} = (24 - 26.333)/2.606 = -0.90 \rightarrow$ random; $a = 38$ gives $Z_{\blacksquare} = +4.48 \rightarrow$ clearly not random.

Write the algorithm for LCM. Using multiplicative congruential method, find the period for $a=13$, $m=26$, $X_0=2$ and 4

Answer

Algorithm for LCM:

1. Choose the seed X_0 , the multiplier a , the increment c and the modulus m (with $X_0, a, c < m$).
2. Compute the next value by $X_{i+1} = (a \times X_i + c) \bmod m$.
3. The normalized random number is $R_{i+1} = X_{i+1}/m$, which always lies in $[0, 1)$.
4. Set $X_i = X_{i+1}$ and repeat from step 2 for as many numbers as needed.
5. If $c \neq 0$ the generator is called mixed congruential; if $c = 0$ it is multiplicative congruential ($X_{i+1} = a X_i \bmod m$).

Multiplicative congruential with $a = 13$, $m = 26$:

$$X_{i+1} = (13 X_i) \bmod 26$$

For $X_0 = 2$:

1. $X_1 = (13 \times 2) \bmod 26 = 26 \bmod 26 = 0$
2. $X_2 = (13 \times 0) \bmod 26 = 0$, and every later value is 0.

Sequence: $2 \rightarrow 0 \rightarrow 0 \rightarrow 0 \dots$ **period = 1** (degenerate).

For $X_0 = 4$:

1. $X_1 = (13 \times 4) \bmod 26 = 52 \bmod 26 = 0$
2. All later values are 0 — **period = 1** (degenerate).

Conclusion: $a = 13$ and $m = 26$ share the common factor 13, so every multiple of 13 vanishes modulo 26 and both seeds collapse to zero. The generator is worthless, which shows the general rule: in a multiplicative congruential generator the multiplier and the modulus must be chosen so that the numbers produced cycle through a long period, and a seed sharing a factor with m destroys the period. In practice m is taken as a large prime (or a large power of two with $a \equiv 1 \pmod{4}$) and m must be larger than the longest run.

2012 C · 2 marks

What is the role of random numbers in simulation?

Answer

The role of random numbers in simulation is to **imitate the chance elements of a real system**. Almost every stochastic simulation needs them to sample from the input distributions of the model:

- Sampling how many customers arrive in an interval (Poisson arrivals)
- Sampling the time between two arrivals (exponential inter-arrival times)
- Sampling service times, machine failure and repair times, demand levels, lead times
- Sampling the discrete events of random processes and, in Monte Carlo work, generating the sample points of a numerical estimate

The numbers are used because every such quantity is defined in terms of a uniform random number on (0, 1), transformed by the inverse-transform, Box-Muller or rejection method into the required distribution. Thus the quality of the whole simulation depends on the quality of the random number stream: if the stream is not uniform and independent, every sampled value is biased and so is the result of the study.

2012 C · 5 marks

Generate up to 5 random numbers using LCM: $X_{\blacksquare} = \text{seed}$, $a=17$, $c=43$, $m=100$

Answer

Formula: $X_{\blacksquare\blacksquare\blacksquare\blacksquare} = (a X_{\blacksquare} + c) \bmod m = (17 X_{\blacksquare} + 43) \bmod 100$. The question gives the seed as $X_{\blacksquare} = \text{seed}$; the standard demonstration of these parameters uses $X_{\blacksquare} = 27$, which is the value worked below.

n	$X_{\blacksquare}$	$17 X_{\blacksquare} + 43$	$X_{\blacksquare\blacksquare\blacksquare\blacksquare} = \text{result} \bmod 100$	$R_{\blacksquare\blacksquare\blacksquare\blacksquare} = X_{\blacksquare\blacksquare\blacksquare\blacksquare}/100$
0	27	$17 \times 27 + 43 = 502$	2	0.02
1	2	$17 \times 2 + 43 = 77$	77	0.77
2	77	$17 \times 77 + 43 = 1352$	52	0.52
3	52	$17 \times 52 + 43 = 927$	27	0.27
4	27	$17 \times 27 + 43 = 502$	2	0.02

Five random numbers: 2, 77, 52, 27, 2, i.e. the normalized values 0.02, 0.77, 0.52, 0.27, 0.02.

The sequence returns to 27 after four steps, so the **period is only 4**. These parameters do not satisfy the full-period conditions, which is why the generator repeats almost immediately.

2012 C · 8 marks

Apply auto-correlation test: check whether 3rd, 8th, 13th and so on numbers are independent. ($Z_{\text{calc}}=1.96$, $\alpha=0.05$). [30 numbers given]

Answer

The **auto-correlation test** detects dependence between numbers a fixed distance apart. Taking the 3rd, 8th, 13th ... values means the lag is $l = 5$, and only those values are used, so the sequence of 30 numbers yields the series $R_3, R_8, R_{13}, R_{18}, R_{23}, R_{28}$.

Formulas

$$\rho_{\text{calc}} = (1/M) \sum R_i R_{i+l} - 0.25 \quad \sigma_{\rho_{\text{calc}}} = \sqrt{(13M + 7)/(12(M + 1))} \quad Z_{\text{calc}} = \rho_{\text{calc}}/\sigma_{\rho_{\text{calc}}}$$

Here $M = 5$ pairs: $(R_3, R_8), (R_8, R_{13}), (R_{13}, R_{18}), (R_{18}, R_{23}), (R_{23}, R_{28})$.

$\sigma_{\rho_{\text{calc}}} = \sqrt{(13 \times 5 + 7)/(12 \times 6)} = \sqrt{72/72} = 8.485/72 = \mathbf{0.1178}$; the critical value is $Z_{\text{crit}} = \mathbf{1.96}$.

Worked illustration of the arithmetic

The same formula on a small self-contained example — 10 numbers with lag $l = 2$, giving $M = 8$ pairs:

Numbers: 0.42, 0.18, 0.65, 0.03, 0.79, 0.55, 0.32, 0.11, 0.88, 0.61

Pair	$R_i \times R_{i+l}$
(0.42, 0.65)	0.2730
(0.18, 0.03)	0.0054
(0.65, 0.79)	0.5135
(0.03, 0.55)	0.0165
(0.79, 0.32)	0.2528
(0.55, 0.11)	0.0605
(0.32, 0.88)	0.2816

(0.11, 0.61)	0.0671
--------------	--------

$$\Sigma = 1.4704, \text{ so } \rho_{\blacksquare} = (1/8)(1.4704) - 0.25 = 0.1838 - 0.25 = \mathbf{-0.0662}$$

$$\sigma_{\rho_{\blacksquare}} = \sqrt{(13 \times 8 + 7)/(12 \times 9)} = \sqrt{111/108} = 10.535/108 = \mathbf{0.0975}$$

$$Z_{\blacksquare} = -0.0662/0.0975 = \mathbf{-0.679}$$

Decision

$|Z_{\blacksquare}| = 0.679 \leq 1.96$, so the null hypothesis of independence is not rejected: the values are **not auto-correlated** — the numbers are independent at the 5% level. The same three steps (form the lag-5 pairs, compute $\rho_{\blacksquare}$, divide by $\sigma_{\rho_{\blacksquare}}$ and compare with 1.96) give the answer for the 30 numbers of the question.

2011 F · 6 marks

Using multiplicative congruential method, find period for $a=13$, $m=64$, $X_{\blacksquare}=1,2,3,4$.

Answer

Formula: $X_{\blacksquare\blacksquare\blacksquare} = (13 \times X_{\blacksquare}) \bmod 64$

For $X_{\blacksquare} = 1$:

$X_{\blacksquare}=13$, $X_{\blacksquare\blacksquare}=13 \times 13 \bmod 64 = 169 \bmod 64 = 41$, $X_{\blacksquare\blacksquare\blacksquare}=13 \times 41 \bmod 64 = 533 \bmod 64 = 21$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 21 \bmod 64 = 273 \bmod 64 = 17$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 17 \bmod 64 = 221 \bmod 64 = 29$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 29 \bmod 64 = 377 \bmod 64 = 57$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 57 \bmod 64 = 741 \bmod 64 = 37$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 37 \bmod 64 = 481 \bmod 64 = 33$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 33 \bmod 64 = 429 \bmod 64 = 45$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 45 \bmod 64 = 585 \bmod 64 = 9$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 9 \bmod 64 = 117 \bmod 64 = 53$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 53 \bmod 64 = 689 \bmod 64 = 49$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 49 \bmod 64 = 637 \bmod 64 = 61$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 61 \bmod 64 = 793 \bmod 64 = 25$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 25 \bmod 64 = 325 \bmod 64 = 5$, $X_{\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare}=13 \times 5 \bmod 64 = 65 \bmod 64 = 1$

Period for $X_{\blacksquare}=1$: **16** (returns to 1 after 16 steps)

For $X_{\blacksquare} = 2$: $X_{\blacksquare}=26$, $X_{\blacksquare\blacksquare}=338 \bmod 64=18$, $X_{\blacksquare\blacksquare\blacksquare}=234 \bmod 64=42$... Period = 16

For $X_{\blacksquare} = 3$: $X_{\blacksquare}=39$, $X_{\blacksquare\blacksquare}=507 \bmod 64=59$, $X_{\blacksquare\blacksquare\blacksquare}=767 \bmod 64=63$... Period = 16

For $X_{\blacksquare} = 4$: $X_{\blacksquare}=52$, $X_{\blacksquare\blacksquare}=676 \bmod 64=36$, $X_{\blacksquare\blacksquare\blacksquare}=468 \bmod 64=20$... Period = 16

All seeds produce **period = 16** (which is $m/4 = 64/4 = 16$).

2011 C · 2+5 marks

What does the Kolmogorov–Smirnov test indicate? Perform the K–S test with data 0.24, 0.89, 0.11, 0.61, 0.23, 0.86, 0.41, 0.64, 0.50, 0.65 ($\alpha = 0.05$).

Answer

What the K–S test indicates: it measures the largest vertical gap between the empirical cumulative distribution $F_n(x)$ of the sample and the theoretical cumulative distribution $F(x)$ of the hypothesised distribution (here the uniform distribution on $[0,1)$, so $F(x) = x$). A small D means the sample could plausibly have come from that distribution; a large D means it could not. Unlike the chi-square test it needs no grouping into intervals, which is why it is considered more powerful.

$$D = \max_i \{ |i/N - R_i|, |R_i - (i-1)/N| \} \quad D_{\text{critical}} = D_{\alpha}/\sqrt{N}$$

Step 1 — sort the 10 numbers

0.11, 0.23, 0.24, 0.41, 0.50, 0.61, 0.64, 0.65, 0.86, 0.89

Step 2 — compute the deviations ($N = 10$)

i	R_i	i/N	$ i/N - R_i $	$(i-1)/N$	$ R_i - (i-1)/N $
1	0.11	0.1	0.01	0.0	0.11
2	0.23	0.2	0.03	0.1	0.13
3	0.24	0.3	0.06	0.2	0.04
4	0.41	0.4	0.01	0.3	0.11
5	0.50	0.5	0.00	0.4	0.10
6	0.61	0.6	0.01	0.5	0.11
7	0.64	0.7	0.06	0.6	0.04
8	0.65	0.8	0.15	0.7	0.05
9	0.86	0.9	0.04	0.8	0.06
10	0.89	1.0	0.11	0.9	0.01

Step 3 — the statistic and the critical value

The largest deviation in the table is **$D = 0.15$** (at $i = 8$).

$D_{\text{critical}} = D_{0.05}/\sqrt{N} = 1.36/\sqrt{10} = 1.36/3.162 = \mathbf{0.430}$ (the exact small-sample table gives ≈ 0.41 for $N = 10$).

Conclusion

Since $0.15 < 0.430$, we **fail to reject H_0** . The sample is statistically consistent with a uniform $[0,1)$ distribution — the numbers can be accepted as random.

2011 C · 2+3 marks

What do you mean by pseudo random number? How do these random numbers help in simulation?

Answer

Pseudo-random number: a number produced by a deterministic algorithm which nevertheless passes the tests for uniformity and independence, so it behaves as if it were random. It is called pseudo because the same seed always regenerates the same sequence.

How they help in simulation:

1. **Artificial sampling:** a simulation has to imitate chance events — how many customers arrive, how long service takes, which machine fails. The generator supplies the random values from which samples of these distributions are drawn (inverse transform, Box-Muller, etc.).
2. **Monte Carlo computations:** the same numbers provide the sampling points that estimate integrals, areas and the value of π .
3. **Reproducibility of experiments:** because the sequence follows from the seed, a run can be repeated exactly. If random results had to be reproduced with physical randomness this would be impossible to do reliably.
4. **Different independent streams:** separate streams can be assigned to different random components (arrivals, service, failures), which improves the statistical quality of the study.
5. **Speed and cost:** they are generated on the computer itself in microseconds, unlike tables of random digits which must be stored and are quickly exhausted.

2011 F · 5 marks

Explain combined linear congruential method for generating random numbers

Answer

A **combined linear congruential generator** runs several congruential generators with different moduli and combines their outputs into a single stream.

For $j = 1, 2, \dots, k$, define the components:

$$X_{j,n+1} = (a_j X_{j,n} + c_j) \bmod m_j$$

Then combine them (L'Ecuyer):

$$X_n = \left(\sum_{j=1..k} (-1)^{j-1} X_{j,n} \right) \bmod (m_1 - 1) + 1 \quad R_n = X_n / m_1$$

(with $R_n = (m_1 - 1)/m_1$ used if $X_n = 0$).

Why it is used

1. **Far longer period.** The period of the combined generator is approximately the product (or least common multiple) of the component periods, so periods of the order of 10^{18} are reached even though each component alone has a period of only 10^9 .
2. **Better statistical quality.** Combining breaks the regular lattice structure that makes the output of a single LCM fall on a small number of hyperplanes, so the combined stream passes uniformity and independence tests that the components fail.
3. **Independent streams.** The components can be advanced separately, giving several disjoint streams that can be assigned to different random components of a simulation (arrivals, service, failures) or to different processors.

Example of parameters (L'Ecuyer, 1988): $m_1 = 2147483563$, $a_1 = 40014$; $m_2 = 2147483399$, $a_2 = 40692$. Both components are multiplicative ($c = 0$) and the combined generator has a period of about 2.3×10^{18} — the classic generator used by many simulation packages.

2011 C · 3 marks

Why do we need to test the randomness of the number in simulation?

Answer

Random numbers must be tested because the generator is a **deterministic formula**, so any weakness in it produces a stream with a hidden pattern, and every such weakness corrupts the simulation:

1. All sampling in a stochastic simulation is derived from uniform random numbers ($x = a + (b - a)R$, inverse transform, Box-Muller). If R is not uniform or not independent, **every sampled variable is biased**, so arrival patterns, service times and hence the whole output of the study are wrong.
2. A generator may have a **much shorter period** than the run length, so the simulation replays the same events and produces an artificially stable, over-optimistic result.
3. A generator may produce values that are **correlated** (the lattice structure of a poor LCM), which makes the observations in the run statistically dependent; the confidence intervals and replication analysis of Chapter 7 then become invalid.

4. Some methods **degenerate** to a constant (mid-square $\rightarrow 0$), which silently stops the simulation from being stochastic at all.
5. The principle is **garbage in, garbage out**: no amount of careful model building or statistical analysis can recover from a bad random number stream, so testing is done before the model is trusted.

The tests used are uniformity tests (frequency, chi-square, Kolmogorov–Smirnov) and independence tests (auto-correlation, run, gap, poker).

2011 C · 3 marks

Explain the significance of chi square test

Answer

The **chi-square test** is a goodness-of-fit test. In simulation it is used for two purposes:

1. To test whether the numbers produced by a generator are **uniformly distributed** on (0, 1) — the uniformity half of the randomness requirement.
2. To test whether an **input distribution fitted to data** (exponential, Poisson, etc.) is a good fit, which is part of validating the model's data assumptions.

Formula:

$$\chi^2 = \sum (O_i - E_i)^2 / E_i$$

where O_i is the observed frequency in interval i , $E_i = N/k$ the expected frequency, N the sample size and k the number of intervals.

Steps: divide (0, 1) into k equal intervals $\rightarrow$ count the observed frequency in each $\rightarrow$ compute $E_i = N/k \rightarrow$ compute $\chi^2 \rightarrow$ compare with the table value $\chi^2(\alpha, k - 1)$. If the calculated χ^2 is less than the table value, the observed distribution does not differ significantly from the uniform, and the numbers are accepted as random.

Significance: it converts a visual impression ("the values look evenly spread") into a quantified decision at a stated confidence level, and it does so for the whole distribution at once. Its limitation is that the result depends on the number of intervals chosen, and it needs a reasonably large sample ($E_i \geq 5$ in every interval) — which is why the Kolmogorov–Smirnov test is often used alongside it.

2011 F · 5 marks

A sequence of 1000 three-digit numbers: 680 have three different digits, 289 contain exactly one pair of like digits, 31 contain three like digits. Are these numbers independent? ($\alpha=0.05$)

Answer

This is the **poker test**, which tests independence by classifying three-digit numbers according to the pattern of their digits.

Theoretical probabilities for three-digit numbers:

- All three digits different: $P = 10 \times 9 \times 8 / 10^3 = 720 / 1000 = \mathbf{0.720}$
- Exactly one pair of like digits: $P = 1 - 0.720 - 0.010 = \mathbf{0.270}$
- All three digits alike: $P = 10 / 10^3 = \mathbf{0.010}$

Expected frequencies for 1000 numbers: 720, 270 and 10.

Category	Observed O_{ij}	Expected E_{ij}	$(O_{ij} - E_{ij})^2 / E_{ij}$
All different	680	720	$(680 - 720)^2 / 720 = 1600 / 720 = 2.222$
Exactly one pair	289	270	$(289 - 270)^2 / 270 = 361 / 270 = 1.337$
Three alike	31	10	$(31 - 10)^2 / 10 = 441 / 10 = 44.100$

$$\chi^2 = 2.222 + 1.337 + 44.100 = 47.659$$

Degrees of freedom = $k - 1 = 3 - 1 = 2$, so at $\alpha = 0.05$ the critical value is $\chi^2(0.05, 2) = 5.991$.

Conclusion: since $47.659 > 5.991$, the null hypothesis of independence is **rejected**. The numbers are **not independent**. The rejection is caused almost entirely by the last category — 31 numbers with three like digits were observed where only 10 were expected — a strong sign that the generator repeats digits far too often. The generator must be replaced (or its parameters corrected) before it is used in a simulation.

2010 C · 10 marks

Write down the steps for the Kolmogorov–Smirnov test.

Answer

The K–S test compares the empirical (sample) cumulative distribution with the theoretical cumulative distribution of the hypothesised distribution.

Steps

1. **State the hypotheses.** H_0 : the numbers follow the theoretical distribution $F(x)$ (for random numbers on $[0,1)$, $F(x) = x$). H_1 : they do not.
2. **Sort the sample** in ascending order: $R_1 \leq R_2 \leq \dots \leq R_N$.
3. **Compute the empirical CDF** at each point: $F_n(R_i) = i/N$, and note that just before R_i it is $(i - 1)/N$.
4. **Compute the theoretical CDF** $F(R_i) = R_i$ for each i .
5. **Compute both deviations** at every i and take the maximum absolute value:
 $D = \max \text{ over } i \text{ of } \{ |i/N - R_i|, |R_i - (i - 1)/N| \}$.
6. **Obtain the critical value** $D_{\text{critical}} = D_{\alpha}/\sqrt{N}$ ($D_{0.05} = 1.36$, $D_{0.01} = 1.63$ for large N ; use the exact table when $N < 35$).
7. **Decide.** If $D \leq D_{\text{critical}}$, fail to reject H_0 — the numbers are uniformly distributed. If $D > D_{\text{critical}}$, reject H_0 — the generator is not producing the required distribution.

Interpretation

D is simply the largest vertical distance between the two cumulative curves, so the test is sensitive to the entire shape of the distribution rather than to a few intervals. It requires no grouping, works for small samples, and can test any hypothesised distribution — but it is only valid when the parameters of $F(x)$ are the true theoretical ones and not estimated from the sample.

2010 C · 3 marks

How pseudo random numbers differ from random numbers

Answer

Aspect	Truly random numbers	Pseudo-random numbers
Source	A physical process — radioactive decay, thermal noise, electronic noise	A deterministic mathematical formula (LCM, mid-square) seeded with a starting value
Reproducibility	Cannot be reproduced; the sequence is lost once generated	Fully reproducible from the seed
Period	No period (in principle), but the source may drift	Definite period; usually very long but finite
Uniformity	Is uniform only if the physical source is well built and unbiased	Uniform inside the period if the parameters satisfy the conditions

Speed / cost	Slow and expensive; needs special hardware	Very fast and cheap; pure arithmetic on the host machine
Predictability	Not predictable	Predictable if the algorithm and seed are known
Use in simulation	Rarely used (only to seed or to verify)	The standard input to every discrete-event simulation and Monte Carlo study

Key point: the difference is the *origin*, not the appearance. A good pseudo-random sequence is indistinguishable from a truly random one by every practical test, and it is preferred in simulation because it can be reproduced exactly.

2010 F · 3 marks

When generating pseudo-random numbers, certain problems or errors can occur. What are they?

Answer

Several characteristic faults can arise when pseudo-random numbers are generated:

1. **Short period / cycling:** the sequence repeats much sooner than required. If the period is shorter than the number of values the simulation needs, the same history is replayed and the results are biased. Example: LCM with $X_0=27$, $a=17$, $c=43$, $m=100$ has period only 4.
2. **Degeneracy (falling to a constant):** the mid-square method reaches 0000 and stays at 0 for ever. The multiplicative LCM with $a=13$, $m=26$ collapses to 0 from any seed, because a and m are not coprime.
3. **Non-uniform distribution:** the generated values cluster instead of filling $(0, 1)$ evenly, so the empirical distribution differs from the uniform distribution the model assumes.
4. **Correlation between successive values (lattice structure):** LCM output lies on a small number of hyperplanes in k dimensions; a poor choice of a , c and m makes this visible as structure in scatter plots.
5. **Poor low-order bits:** in a straightforward LCM the least significant digits are much less random than the high-order digits, so the generator must not be used by taking only those digits.
6. **Bad seed selection:** seeds such as 0 (for a multiplicative generator) or values outside the period produce a trivial or degraded sequence.
7. **Non-random dependence on the modulus:** choosing m as a power of two makes the period and the pattern of the low-order bits weak; m should be a large prime or a large power of a prime with $a \equiv 1 \pmod 4$.
8. **Implementation errors:** integer overflow and precision loss when the arithmetic is done in floating point produce incorrect, repeated or biased values.

These faults are why no generator may be used in a study without first passing the battery of statistical tests described in sections 6.7–6.13.

2010 C · 3+4 marks

Using multiplicative congruential method, find the period and generate for $a=17$, $m=64$, $X_0=3$

Answer

Multiplicative congruential generator: $X_{n+1} = (a X_n) \bmod m = (17 X_n) \bmod 64$, with $X_0 = 3$.

n	X_n	$17 X_n$	$X_{n+1} = \text{result} \bmod 64$	$R_{n+1} = X_{n+1}/64$
0	3	51	51	0.7969
1	51	867	$867 - 832 = 35$	0.5469
2	35	595	$595 - 576 = 19$	0.2969
3	19	323	$323 - 320 = 3$	0.0469

Generated numbers: 51, 35, 19, 3 — normalized **0.7969, 0.5469, 0.2969, 0.0469**.

The fourth value returns to the seed 3, so the **period is 4**. The period is short because $m = 64$ is a power of two and the seed 3 is odd and coprime to m , so the cycle is limited to the small set of values reachable from 3. This is the practical objection to multiplicative generators with a power-of-two modulus: period $m/4$ at best for this kind of parameter set, rather than the maximum $m - 1$.

2010 F · 7 marks

Use multiplicative congruential method to generate four three-digit random numbers. $x_0=117$, $a=43$, $m=1000$

Answer

Multiplicative congruential generator: $X_{n+1} = (a X_n) \bmod m = (43 X_n) \bmod 1000$, with $x_0 = 117$. Since $m = 1000$, every generated value is written as a **three-digit** number (leading zeros kept).

n	X_n	$43 \times X_n$	$X_{n+1} = \text{result} \bmod 1000$	Three-digit form	$R_{n+1} = X_{n+1}/1000$
0	117	$43 \times 117 = 5031$	$5031 - 5000 = 31$	031	0.031

1	31	$43 \times 31 = 1333$	$1333 - 1000 =$ 333	333	0.333
2	333	$43 \times 333 =$ 14319	$14319 - 14000$ $=$ 319	319	0.319
3	319	$43 \times 319 =$ 13717	$13717 - 13000$ $=$ 717	717	0.717

Four three-digit random numbers: 031, 333, 319, 717, i.e. the uniform values **0.031, 0.333, 0.319, 0.717**.

Note that the seed 117 itself is not part of the output — the generator produces the values that follow it. Also note the leading zero in 031: discarding it would destroy the uniformity of the low-order digits.

2010 C · 10 marks

Test auto-correlation of 2nd, 9th, 16th, ... numbers. ($\alpha=0.05$). [40 numbers given]

Answer

Taking the 2nd, 9th, 16th ... values means the lag is $l = 7$. From a sequence of 40 numbers the series is $R_2, R_9, R_{16}, R_{23}, R_{30}, R_{37}$, giving the $M = 5$ pairs (2,9), (9,16), (16,23), (23,30), (30,37).

Formulas and constants

$$\rho_{ll} = (1/M) \sum R_{l_i} R_{l_{i+1}} - 0.25 \quad \sigma_{\rho_{ll}} = \sqrt{(13M + 7)/(12(M + 1))} \quad Z_{ll} = \rho_{ll}/\sigma_{\rho_{ll}}$$

$$\text{With } M = 5: \sigma_{\rho_{ll}} = \sqrt{(13 \times 5 + 7)/(12 \times 6)} = \sqrt{72/72} = \mathbf{0.1178}$$

At $\alpha = 0.05$ the two-tailed critical value is $Z_{ll,0.025} = \mathbf{1.96}$.

Procedure

- Write down the five selected values in order.
- Multiply each value by the next one in the series (the value one lag later) and add the five products: $S = \sum R_{l_i} R_{l_{i+1}}$.
- $\rho_{ll} = S/M - 0.25$.
- $Z_{ll} = \rho_{ll}/0.1178$.
- Compare with ± 1.96 .

Interpretation of the possible outcomes

Computed Z_{AC}	Conclusion
$ Z_{\text{AC}} \leq 1.96$	Fail to reject H_0 — the numbers are independent, no auto-correlation at lag 7.
$Z_{\text{AC}} > 1.96$	Reject H_0 — the values a lag apart tend to rise and fall together (positive auto-correlation).
$Z_{\text{AC}} < -1.96$	Reject H_0 — the values alternate (negative auto-correlation).

Numerical meaning of ρ_{AC} : ρ_{AC} measures the correlation between the value at position i and the value at position $i + 7$. It is compared against the theoretical value 0.25 (the expected product of two independent uniform values) rather than against 0, which is why 0.25 is subtracted in the formula. A value of ρ_{AC} close to zero after this correction means no correlation; a large positive ρ_{AC} means the generator carries a hidden trend across the sequence, which would invalidate the independence assumption of every replication in the study.

For a 10-mark answer, state the formulas, show the five products in a table, compute ρ_{AC} and $\sigma_{\rho_{\text{AC}}}$, obtain Z_{AC} , and finish with the comparison against 1.96 and the written conclusion.

Chapter 7 — Analysis of Simulation Output (6 marks)

7.1 Estimation Methods

Simulation outputs are **random variables** because inputs contain randomness. Therefore, statistical analysis is essential to draw valid conclusions.

Why Analyze Simulation Output?

Without proper statistical analysis, simulation results may be misleading. We need to determine the accuracy and reliability of results.

Types of Simulations

Terminating Simulation	Non-Terminating (Steady-State) Simulation
Has a natural ending point	No natural ending point
Example: Bank closes at 5 PM	Example: 24/7 emergency room
Analyze each run independently	Need to handle initial bias

Point Estimation

A single value estimate of a parameter:

$$\bar{X} = (1/n) \times \sum X_i \text{ (Sample Mean)}$$

The sample mean $\bar{X}$ is the point estimate of the population mean μ .

Interval Estimation (Confidence Intervals)

A range of values likely to contain the true parameter:

$$\bar{X} \pm t_{\{\alpha/2, n-1\}} \times S/\sqrt{n}$$

where S = sample standard deviation, n = number of observations, t = t-distribution critical value

Confidence interval for the mean waiting time 5 replications · 95% · $t = 2.776$

Observations **4.2, 3.8, 5.1, 4.5, 3.9 min**

$n = 5$

Degrees of freedom $n - 1 = 4$

1. $\bar{X} = (4.2 + 3.8 + 5.1 + 4.5 + 3.9) / 5 = 21.5 / 5 = 4.30 \text{ min}$

2. $S^2 = [(4.2-4.3)^2 + (3.8-4.3)^2 + (5.1-4.3)^2 + (4.5-4.3)^2 + (3.9-4.3)^2] / (5-1) = [0.01 + 0.25 + 0.64 + 0.04 + 0.16] / 4 = 1.10 / 4 = 0.275$
Divide by $n - 1 = 4$, not by $n = 5$: a sample variance uses the degrees of freedom.

3. $S = \sqrt{0.275} = 0.5244$

4. $95\% \text{ CI} = \bar{X} \pm t \times S/\sqrt{n} = 4.30 \pm 2.776 \times 0.5244/\sqrt{5} = 4.30 \pm 2.776 \times 0.2345 = 4.30 \pm 0.651$

95% CI **(3.65, 4.95) minutes**

Common mistake

An earlier draft of this example divided the squared deviations by n and came out with $S = 0.508$ and the interval $(3.67, 4.93)$. Both numbers were wrong. The variance always divides by $n - 1$, and the standard error is $S/\sqrt{n}$ — not S alone.

7.2 Simulation Run Statistics

Measures collected during simulation runs:

- **Mean:** Average of the output measure
- **Variance:** Spread of results around the mean
- **Minimum & Maximum:** Range of observed values
- **Utilization:** Fraction of time a resource is busy
- **Throughput:** Number of entities processed per unit time

Types of Statistics

Time-Averaged Statistics	Observation-Based Statistics
Averaged over simulated time	Based on individual observations
Example: Average queue length over 8 hours	Example: Average waiting time per customer
Weighted by time duration	Simple average of observations

Batch Means Method

For long single runs: divide the run into **batches** of equal length. Treat each batch mean as an independent observation. Use batch means to calculate confidence intervals.

7.3 Replication of Runs

Most Frequently Asked Topic (~10 times in past exams!)

Replication of runs is the MOST important topic in this chapter.

Replication means running the simulation **multiple times** with **different random number seeds**. Each run is called a *replication* and is statistically independent.

Why Replication is Necessary

1. **Get independent observations:** Each replication provides an independent data point
2. **Estimate variability:** Multiple runs show how much results vary
3. **Calculate confidence intervals:** Need multiple observations for statistical inference
4. **Reduce sampling error:** More replications → more accurate estimates
5. **A single run may give misleading results** due to random variation

Mathematical Modeling

Let $Y_1, Y_2, \dots, Y_n$ be outputs of n **independent replications**. These are **IID** (Independent and Identically Distributed) random variables.

Point Estimate: $\bar{Y} = (1/n) \times \sum Y_i$

Variance Estimate: $S^2 = (1/(n-1)) \times \sum (Y_i - \bar{Y})^2$

Confidence Interval: $\bar{Y} \pm t_{\{\alpha/2, n-1\}} \times S/\sqrt{n}$

Replication of runs — point estimate and confidence interval
 $n = 5 \cdot 95\% \cdot t = 2.776$

Average wait Y_i **12.5, 14.2, 11.8, 13.6, 12.9 min**

Replications $n = 5$

Degrees of freedom $n - 1 = 4$

1. $\bar{Y} = (12.5 + 14.2 + 11.8 + 13.6 + 12.9) / 5 = 65.0 / 5 = 13.0$ min
This is the point estimate of the true mean — one number from five independent runs.
2. $S^2 = [0.25 + 1.44 + 1.44 + 0.36 + 0.01] / 4 = 3.50 / 4 = 0.875$
Squaring each $(Y_i - 13.0)$: $-0.5, +1.2, -1.2, +0.6, -0.1$.
3. $S = \sqrt{0.875} = 0.9354$ min
4. $95\% \text{ CI} = \bar{Y} \pm t \times S/\sqrt{n} = 13.0 \pm 2.776 \times 0.9354/2.2361 = 13.0 \pm 2.776 \times 0.4183 = 13.0 \pm 1.161$
If this interval is too wide to decide anything, the answer is more replications — the width shrinks as $1/\sqrt{n}$.

$\bar{Y} = 13.0$ min **95% CI (11.84, 14.16) minutes**

7.4 Elimination of Initial Bias

At the start of simulation, the system is typically in an **unrealistic state** (empty and idle). These initial observations are **biased** and don't represent steady-state behavior.

The Warm-Up Problem

When a simulation starts from an empty state (e.g., bank with no customers at $t=0$), the initial period produces observations that are not representative of normal operation. This is called **initial transient** or **initialization bias**.

Methods to Handle Initial Bias

1. **Long Run Approach:** Run the simulation long enough that initial bias becomes negligible compared to the total run length.
Truncation (Deletion): Delete initial observations. Use **Welch's method** to determine the truncation point:
 - Run multiple replications
 - Calculate moving averages across replications
 - Plot the moving average — the deletion point d is where it appears to converge to steady state
3. **Moving Average Approach:** Use moving averages to identify when steady state begins. The point where the moving average stabilizes is the warm-up period.
4. **Start in Busy State:** Initialize the simulation in a state that represents typical operation (e.g., some customers already in queue) rather than empty.

Welch's Method

Make n replications, each of length m . Calculate the ensemble average $\bar{X}(t)$ at each time t . Apply moving average of window w . Plot $\bar{X}_w(t)$ and choose deletion point d where it appears to converge.

Past questions (6 of 6)

2015 F · 5 marks · repeated 8x

What is initial bias in simulation output? Mention the idea of elimination of initial bias.

Answer

Initial Bias: When a simulation starts from an empty/idle state (e.g., bank with no customers at time 0), the early observations do not represent the system's normal (steady-state) operation. These biased initial observations are called the **initial transient** or **warm-up period**.

Example: In a bank simulation starting from empty, the first few customers have zero waiting time. But in reality, the bank is rarely empty during operating hours.

Methods for Elimination

1. **Long Run:** Run simulation long enough that initial bias is negligible relative to total data
2. **Truncation:** Delete the first d observations. Use Welch's method (moving average across replications) to determine d
3. **Moving Average:** Plot moving averages to visually identify when steady state begins
4. **Initialized Start:** Start simulation in a busy/typical state rather than empty

2015 F · 5 marks · repeated 6x

Explain the mathematical modeling of replication of runs.

Answer

Let $Y_1, Y_2, \dots, Y_n$ be outputs of **n independent replications** of a simulation. Each Y_i is obtained by running the simulation with a different random number seed.

These $Y_1, Y_2, \dots, Y_n$ are **IID** (Independent and Identically Distributed) random variables.

Point Estimate (Sample Mean):

$$\bar{Y} = (1/n) \times \sum_{i=1}^n Y_i$$

Variance Estimate:

$$S^2 = (1/(n-1)) \times \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Confidence Interval:

$$\bar{Y} \pm t_{\{\alpha/2, n-1\}} \times S/\sqrt{n}$$

As n increases, $S/\sqrt{n}$ decreases, giving narrower (more precise) confidence intervals. The choice of n depends on the desired precision and available computing resources.

2019 F · 10 marks · repeated 5x

'Simulation output has to be analyzed.' Justify the statement. Explain replication of runs and elimination of initial bias.

Answer

Why Simulation Output Must Be Analyzed

Simulation outputs are **random variables** because the inputs include random components. A single simulation run produces only estimates, not exact values. Without statistical analysis:

- Results may be misleading due to random variation
- We cannot determine the accuracy or reliability of results
- Decision makers cannot know how confident they should be in the findings

Replication of Runs

Running the simulation **n times with different random seeds**. Each run is independent.

Let $Y_1, Y_2, \dots, Y_n$ be outputs of n replications (IID random variables):

- Point estimate: $\bar{Y} = (1/n) \times \sum Y_i$
- Variance: $S^2 = (1/(n-1)) \times \sum (Y_i - \bar{Y})^2$
- 95% CI: $\bar{Y} \pm t_{\{0.025, n-1\}} \times S/\sqrt{n}$

Purpose: Get independent observations, estimate variability, calculate confidence intervals.

Elimination of Initial Bias

At simulation start, the system is in an unrealistic state (empty/idle). Methods:

1. **Long run:** Run long enough that initial bias becomes negligible
2. **Truncation:** Delete initial observations (use Welch's method to find deletion point)
3. **Moving average:** Identify when steady state begins
4. **Start in busy state:** Initialize with typical operating conditions

2014 F, 2010 F, 2010 C · 4 marks · repeated 5x

Why the output of simulation has to be analyzed?

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 3 marks · repeated 2x

Explain estimation method for analysis of simulation output

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 5 marks

Simulation Run Statistics (Short Note)

Answer

Simulation run statistics are the measures collected during and after simulation runs to evaluate system performance:

- **Mean (Average):** Central tendency of the output measure (e.g., average wait time)
- **Variance/Standard Deviation:** Spread of results around the mean
- **Minimum & Maximum:** Range of observed values
- **Utilization:** Fraction of time a resource (server) is busy
- **Throughput:** Number of entities processed per unit time

Two types:

- **Time-averaged:** Weighted by time duration (e.g., average queue length over 8 hours)
- **Observation-based:** Simple average of individual observations (e.g., average wait per customer)

These statistics are used with confidence intervals from replications to make decisions about the system.

8.1 Basic Concept of Simulation Tools

Simulation tools are software packages designed specifically for building and running simulation models. They reduce programming effort and provide built-in features for model development.

Why Use Simulation Tools?

- **Reduced programming effort:** Pre-built components for common operations
- **Built-in statistical analysis:** Automatic data collection and reporting
- **Standardized model building:** Consistent methodology
- **Faster development:** Less time from concept to working model
- **Automatic event scheduling:** Event management handled by the system
- **Animation and visualization:** Visual representation of model execution

Types of Simulation Languages

General-Purpose Languages	Special-Purpose Simulation Languages
C, C++, Java, Python, FORTRAN	GPSS, SIMSCRIPT, SLAM, Arena, CSSL
Flexible but requires more coding	Built-in simulation constructs
No built-in simulation support	Automatic event scheduling, statistics
Steep learning curve for simulation	Easier for simulation-specific tasks

8.2 CSSLs and GPSS

CSSL (Continuous System Simulation Language)

A family of languages designed for **continuous system simulation** using differential equations.

Examples: CSMP (Continuous System Modeling Program), CSSL-IV, ACSL

CSSL Program Structure

Section	Purpose
INITIAL	Set initial values of variables, define parameters
DYNAMIC	Define the differential equations and system behavior over time
TERMINAL	Post-simulation calculations, output results, clean up

GPSS (General Purpose Simulation System)

Most Important Topic in This Chapter

GPSS is asked in almost every exam paper! Know the block diagram symbols and how to model systems.

GPSS is a **block-oriented language** designed for **discrete-event simulation**. It uses the concept of **transactions** (temporary entities) that flow through **blocks**.

Key Concepts

- **Transactions:** Temporary entities that move through the system (like customers)
- **Blocks:** Operations performed on transactions (like service stations)
- **Facilities:** Resources that can serve one transaction at a time
- **Storages:** Resources that can serve multiple transactions simultaneously

GPSS Block Diagram Symbols

Block	Symbol Shape	Function
GENERATE	Triangle (▴)	Create transactions (arrivals). Parameters: mean inter-arrival time, spread
TERMINATE	Circle (●)	Destroy transactions (departures). Decrements termination counter
SEIZE	Rectangle with notch	Capture a facility (server). Transaction "owns" the facility
RELEASE	Rectangle with notch (opposite)	Free a facility for other transactions

ADVANCE	Rectangle	Delay transaction for specified time (service time)
QUEUE	Rectangle	Enter a queue — records queue statistics
DEPART	Rectangle	Leave a queue — stops recording wait time
ENTER	Rectangle	Enter a storage (multi-server facility)
LEAVE	Rectangle	Leave a storage
TRANSFER	Diamond	Route transactions to different blocks (branching)
TEST	Diamond	Conditional test — route based on condition
START	Control	Begin simulation, set termination counter

GPSS Model: Simple Bank Queue

Customers arrive → wait in queue → get served by teller → leave

Block Diagram:

1. **GENERATE** 5,2 → Create customers (mean=5 min, spread= ± 2 min)
2. **QUEUE** TLINE → Enter the teller line
3. **SEIZE** TELLER → Capture the teller
4. **DEPART** TLINE → Leave the queue
5. **ADVANCE** 4,1 → Service time (mean=4 min, spread= ± 1 min)
6. **RELEASE** TELLER → Free the teller
7. **TERMINATE** 1 → Customer leaves (decrement counter by 1)
8. **START** 100 → Simulate 100 customers

8.3 Discrete Systems Modeling and Simulation

In **discrete-event simulation**, state changes occur only at discrete points in time (events).

Components of Discrete-Event Simulation

- **System State:** Variables describing the system at any time
- **Simulation Clock:** Current simulated time
- **Event List:** List of future events scheduled to occur
- **Statistical Counters:** Accumulate statistics about system performance
- **Initialization Routine:** Set initial values
- **Timing Routine:** Determine next event from event list
- **Event Routines:** Process each type of event
- **Report Generator:** Calculate and output statistics

Clock Advancing Approaches

Next-Event Time Advance	Fixed-Increment Time Advance
Clock jumps to time of next scheduled event	Clock advances by fixed Δt each step
Skips idle periods — more efficient	Checks for events at each increment
Variable time steps	Fixed time steps
Most commonly used	Simpler but less efficient
Used in GPSS, SIMSCRIPT	Used when many events occur continuously

8.4 Continuous Systems Modeling and Simulation

Continuous simulation languages (CSSLs) model systems where state changes continuously over time. Models are specified as sets of **differential equations**.

Numerical methods (Euler, Runge-Kutta) are used to solve the equations at each time step.

CSMP Example: For $dx/dt = -2x$ with $x(0)=10$:

- INITIAL: $X = 10.0$
- DYNAMIC: $\dot{X} = -2.0 \cdot X$, $X = \text{INTGRL}(10.0, \dot{X})$
- TERMINAL: PRINT X

8.5 Structural, Data and Control Statements

SIMSCRIPT

An **event-oriented** simulation language with three main concepts:

- **Entities:** Objects in the system (permanent and temporary)
- **Attributes:** Properties of entities
- **Sets:** Collections of entities (like queues)

Program Organization

Section	Purpose
Preamble	Declarations — define entities, attributes, sets, events
Main	Initialization — set initial conditions, schedule first events
Event Routines	Define what happens when each event occurs

Statement Types

- **Structural Statements:** Define system structure (entities, resources, processes)
- **Data Statements:** Define input data, parameters, distributions
- **Control Statements:** Control simulation execution (START, STOP, RESET, CLEAR)

8.6 Feedback Systems: Typical Applications

Feedback systems in simulation language context involve modeling systems where output influences input.

Applications

- **Home Heating:** Thermostat measures temperature → controls furnace → temperature changes → thermostat adjusts (negative feedback loop)
- **Aircraft Autopilot:** Gyroscope detects heading error → adjusts control surfaces → aircraft changes heading → new error measured
- **Industrial Process Control:** Sensor measures process variable → controller adjusts actuator → process changes → sensor remeasures
- **Economic Models:** Price increase → demand decrease → supply adjustment → new price (market feedback)

Distributed Lag Model

A model where the output depends on current AND past inputs:

$$Y(t) = \sum w_i \times X(t-i) \text{ for } i = 0 \text{ to } k$$

where w_i are weights and $X(t-i)$ are past inputs. Used in economics (effect of investment on GDP) and engineering (thermal systems with delay).

Past questions (15 of 15)

2025 M · 4 marks · repeated 4x

What are the key features of GPSS as a simulation language?

Answer

Key Features of GPSS:

1. **Block-oriented:** Models are built using interconnected blocks that represent operations
2. **Transaction-based:** Temporary entities (transactions) flow through blocks, representing customers, parts, messages
3. **Built-in facilities and storages:** Pre-defined server resources (single-server facilities, multi-server storages)
4. **Automatic statistics:** Queue lengths, utilizations, wait times are automatically tracked
5. **Visual block diagrams:** Each block has a standard symbol (triangle for GENERATE, circle for TERMINATE, etc.)

6. **Event scheduling:** Automatic next-event time advance
7. **Easy to learn:** High-level abstraction makes modeling intuitive

2014 F · 10 marks · repeated 4x

What do you mean by discrete event simulation? Explain principal approaches for advancing simulation clock.

Answer

Discrete-Event Simulation

A type of simulation where the system **state changes only at discrete points in time** called events. Between events, the system state remains unchanged. The simulation tracks individual events (arrivals, departures, failures) and processes them chronologically.

Components: System state, simulation clock, event list, statistical counters, initialization routine, timing routine, event routines, report generator.

Two Approaches for Advancing Simulation Clock

Next-Event Time Advance	Fixed-Increment Time Advance
Clock jumps to the time of next scheduled event	Clock advances by fixed amount Δt each step
Skips idle periods between events	Checks for events at every increment
Variable time steps	Constant time steps
More efficient — most commonly used	Simpler but potentially wasteful
Used in GPSS, SIMSCRIPT, Arena	Used when events occur frequently

Next-Event Process:

1. Scan event list for event with smallest time
2. Advance clock to that event's time
3. Execute the event routine
4. Update statistics and schedule new events
5. Repeat until termination condition

2011 F · 2+4 marks · repeated 4x

What do you mean by Discrete-event simulation? Explain the components and organization of a Discrete event simulation model with flow chart

Practice question — model answer not written yet. Attempt it from the chapter notes.

2015 F · 5 marks · repeated 3x

Draw different types of GPSS block-diagram symbols and explain with GPSS block-diagram of manufacturing shop.

Answer

GPSS Block Symbols

Block	Symbol	Function
GENERATE	■ (Triangle)	Create transactions
TERMINATE	■ (Circle)	Destroy transactions
SEIZE	■ (Rect with left notch)	Capture facility
RELEASE	■ (Rect with right notch)	Free facility
ADVANCE	■ (Rectangle)	Delay time
QUEUE	■ (Rectangle)	Enter queue
DEPART	■ (Rectangle)	Leave queue
TRANSFER	■ (Diamond)	Route/branch

Manufacturing Shop Model

1. **GENERATE** 8,3 — Parts arrive every 8 ± 3 minutes
2. **QUEUE** MACHQ — Enter machine queue
3. **SEIZE** MACHINE — Capture the machine
4. **DEPART** MACHQ — Leave queue
5. **ADVANCE** 6,2 — Machining takes 6 ± 2 minutes

6. **RELEASE MACHINE** — Free the machine
7. **QUEUE INSPQ** — Enter inspection queue
8. **SEIZE INSPECTOR** — Capture inspector
9. **DEPART INSPQ** — Leave inspection queue
10. **ADVANCE 2,1** — Inspection takes 2 ± 1 minutes
11. **RELEASE INSPECTOR** — Free inspector
12. **TERMINATE 1** — Part exits the shop

2025 M · 8 marks

Model a discrete-event system of hospital patient flow using GPSS. Outline key control statements.

Answer

Hospital Patient Flow Model

System: Patients arrive → register → wait → see doctor → get treatment → leave

GPSS Block Diagram

1. **GENERATE 15,5** — Patients arrive every 15 ± 5 minutes
2. **QUEUE REGLINE** — Enter registration queue
3. **SEIZE CLERK** — Capture registration clerk
4. **DEPART REGLINE** — Leave registration queue
5. **ADVANCE 3,1** — Registration takes 3 ± 1 minutes
6. **RELEASE CLERK** — Free the clerk
7. **QUEUE DOCLINE** — Enter doctor's waiting queue
8. **SEIZE DOCTOR** — See the doctor
9. **DEPART DOCLINE** — Leave doctor's queue

10. **ADVANCE** 10,3 — Consultation takes 10 ± 3 minutes
11. **RELEASE** DOCTOR — Free the doctor
12. **ADVANCE** 5,2 — Treatment/pharmacy time 5 ± 2 minutes
13. **TERMINATE** 1 — Patient leaves hospital
14. **START** 100 — Simulate 100 patients

Key Control Statements

- **START n:** Begin simulation, terminate after n transactions complete
- **SIMULATE:** Declares the program as a simulation model
- **RESET:** Clear statistics but continue simulation
- **CLEAR:** Reset entire simulation to initial conditions
- **END:** End of program

2019 F · 10 marks

List out the various simulation tools. Explain GPSS in brief with suitable example.

Answer

Simulation Tools

- **Discrete-Event:** GPSS, SIMSCRIPT, SLAM, Arena, Simul8
- **Continuous:** CSMP, CSSL-IV, ACSL, Simulink
- **General-Purpose:** C/C++, Java, Python, FORTRAN
- **Modern Tools:** AnyLogic, FlexSim, ProModel

GPSS (General Purpose Simulation System)

Block-oriented language for discrete-event simulation. Transactions flow through blocks.

Key Blocks: GENERATE (create), TERMINATE (destroy), SEIZE/RELEASE (use server), ADVANCE (delay), QUEUE/DEPART (queue tracking)

Example: Bank Queue Model

1. **GENERATE** 5,2 — Customers arrive every 5 ± 2 minutes
2. **QUEUE** WAIT — Enter waiting line
3. **SEIZE** TELLER — Capture the teller
4. **DEPART** WAIT — Leave the queue
5. **ADVANCE** 4,1 — Service takes 4 ± 1 minutes
6. **RELEASE** TELLER — Free the teller
7. **TERMINATE** 1 — Customer departs
8. **START** 200 — Simulate 200 customers

GPSS automatically collects statistics: average queue length, maximum queue, server utilization, average wait time.

2015 F · 5 marks

State some types of Simulation languages

Practice question — model answer not written yet. Attempt it from the chapter notes.

2014 F, 2010 F · 3 marks

Why is simulation tool required?

Practice question — model answer not written yet. Attempt it from the chapter notes.

2012 C · 5 marks

Design a simulation model of any discrete system in GPSS symbols

Practice question — model answer not written yet. Attempt it from the chapter notes.

2011 F · 10 marks

What do you mean by distributed lag model? Explain.

Answer

A **distributed lag model** is a model where the output at any time depends not only on the current input but also on **past inputs**. The effect of an input is "distributed" over several time periods.

$$Y(t) = w_0X(t) + w_1X(t-1) + w_2X(t-2) + \dots + w_kX(t-k) = \sum_{i=0}^k w_i \times X(t-i)$$

where:

- $Y(t)$ = output at time t
- $X(t-i)$ = input at time $(t-i)$
- w_i = weight (lag coefficient) for the i -th lag
- k = maximum lag

Applications

- **Economics:** Effect of investment on GDP takes multiple quarters to fully materialize
- **Engineering:** Thermal systems where temperature change lags behind heat input
- **Marketing:** Advertising effect on sales persists over multiple periods

Example

If investment at time t affects GDP at t , $t+1$, and $t+2$ with weights 0.5, 0.3, 0.2:

$$GDP(t) = 0.5 \times \text{Investment}(t) + 0.3 \times \text{Investment}(t-1) + 0.2 \times \text{Investment}(t-2)$$

The total effect of one unit of investment = $0.5+0.3+0.2 = 1.0$ (spread over 3 periods).

2011 C · 2 marks

Show the GPSS block diagram to represent soap quality testing system

Practice question — model answer not written yet. Attempt it from the chapter notes.

2011 F · 6 marks

Discuss SIMSCRIPT and GPSS language in brief

Practice question — model answer not written yet. Attempt it from the chapter notes.

2010 C · 5 marks

CSSL (Short Note)

Answer

CSSL (Continuous System Simulation Language) is a family of languages designed for continuous system simulation using differential equations.

Examples: CSMP (Continuous System Modeling Program), CSSL-IV, ACSL

Program Structure:

- **INITIAL Section:** Set initial values, define parameters and constants
- **DYNAMIC Section:** Define the differential equations and system behavior. Uses INTGRL function for integration.
- **TERMINAL Section:** Post-simulation calculations, output results

Key Features:

- Uses differential equations to describe system behavior
- Automatic numerical integration (Euler, Runge-Kutta methods)
- Suitable for systems like heat transfer, fluid flow, mechanical vibrations
- Provides plotting and tabulation of results

Example: For $dx/dt = -2x$, $x(0) = 10$:

INITIAL: $X = 10.0$

DYNAMIC: $XDOT = -2.0 * X$; $X = \text{INTGRL}(10.0, XDOT)$

TERMINAL: PRINT X

2010 F · 10 marks

Explain Discrete Simulation Language. Explain SIMSCRIPT system concepts and program organization.

Answer

Discrete Simulation Languages

Languages designed specifically for discrete-event simulation where state changes at discrete time points. They provide built-in event scheduling, entity management, and statistical collection.

Examples: GPSS, SIMSCRIPT, SLAM II, SIMAN

SIMSCRIPT

An event-oriented discrete simulation language developed at RAND Corporation.

System Concepts:

Entities: Objects in the system

- *Permanent entities:* Exist throughout simulation (servers, machines)
- *Temporary entities:* Created and destroyed (customers, jobs)

- **Attributes:** Properties of entities (arrival time, priority, service time)

- **Sets:** Collections of entities ordered by some rule (like queues — FIFO, LIFO, ranked)

Program Organization:

Section	Purpose
Preamble	Declarations — define all entities, attributes, sets, and event types. Describes system structure.
Main	Initialization — create permanent entities, set initial values, schedule first events, start simulation
Event Routines	Define processing logic for each event type. When an event occurs, its routine is called to update system state.

Advantages: English-like syntax, flexible entity management, powerful set operations, built-in statistics.

Define state descriptor and discrete event. How do you simulate a telephone system, explain with its states

Practice question — model answer not written yet. Attempt it from the chapter notes.